The Regulatory Intermediation Gap: Evidence from Singapore's Equity Market Reform

Kenneth Khoo*
kenneth.khoo@nus.edu.sg

Hans Tjio†
lawtjioh@nus.edu.sg

Abstract

When governments channel equity market interventions through private intermediaries, the institutional architecture of intermediation shapes how those interventions are priced, systematically favouring firms that already meet intermediary selection criteria over the intended beneficiaries. We term this divergence the *regulatory intermediation gap*. Using a cross-sectional event study of firms listed on the Singapore Exchange (SGX), we examine Singapore's 2025 equity market reform (a S$5 billion capital deployment via professional fund managers, targeted at small- and mid-cap firms, bundled with deregulation of SGX's post-listing oversight regime) and document two dimensions of this gap. First, the market priced capital deployment as a large-cap event: the smallest quartile lost 2.5 percent while the largest gained 0.2 percent, consistent with fund managers' investability constraints that effectively exclude small firms. When mandate terms requiring small- and mid-cap allocation were later disclosed, the size gradient reversed, indicating that precise mandate design can narrow the expected gap. Second, the market penalised firms with accumulated regulatory friction (prior halts and suspensions), interpreting the proposed shift from public to private oversight as information loss — though for financially distressed firms on the watch-list, deregulation was priced as stigma relief. Both findings are instances of a single phenomenon: policy transmitted through institutional channels is priced according to those channels' constraints. The results inform the design of government-intermediated equity programmes in Singapore and comparable initiatives in Japan, Korea, and the European Union.

*Assistant Professor, National University of Singapore, Faculty of Law. We thank the EW Barker Centre for Law & Business for its generous financial support, and Jovan Poh for excellent research assistance.

†CJ Koh Professor of Law, National University of Singapore, Faculty of Law.

1 Introduction

In February 2025, Singapore's Monetary Authority (MAS) announced one of the most ambitious equity market reform packages in the city-state's history. The package deployed two instruments toward a single objective: revitalising an equity market where small- and mid-cap firms had long suffered from thin trading and poor valuations. The first was capital: S$5 billion channelled through professional fund managers under the Equity Market Development Programme (EQDP), intended to bring new institutional money into Singapore-listed equities beyond the index constituents. The second was deregulation: a decisive shift towards a disclosure-based listing regime on the Singapore Exchange (SGX). This included the proposed removal of SGX's financial watch-list for poorly performing companies, consolidation of listing review under a single regulator, and streamlined prospectus requirements, collectively designed to reduce the cost of being listed.

The market's response revealed a striking paradox. Rather than rewarding the small- and mid-cap firms the EQDP was designed to benefit, the announcement generated a size-dependent cross-sectional pattern in which the smallest quartile of equities experienced cumulative abnormal returns of −2.5 percent while the largest quartile gained +0.2 percent, a statistically significant spread of +2.8 percentage points. The market priced the government's equity intervention as a large-cap event. Simultaneously, the reform's deregulatory component penalised firms along a different cross-sectional dimension. Firms with accumulated regulatory friction (prior queries, trading halts, and suspensions) experienced more negative abnormal returns, each additional friction event predicting roughly one-third of a percentage point decline even after controlling for size. The market interpreted the proposed dismantling of SGX's oversight apparatus not as lifting regulatory burden from constrained firms, but as removing informative monitoring from firms that warranted it.[1] Both instruments were thus expected to miss their intended beneficiaries, but for structurally distinct reasons: the capital through *intermediation*, the deregulation through *information loss*.

We define the *regulatory intermediation gap* as the distance between regulatory intent and market outcome when government channels equity market interventions through private intermediaries. Fund managers operating under the EQDP face standard institutional constraints—fiduciary duties, fee structures linked to assets under management (AUM), liquidity requirements for position sizing, and benchmark-relative performance evaluation—that rationally favour large, liquid, index-proximate stocks regardless of stated policy objectives. The announcement-return evidence is consistent with the market pricing this gap as structural: arising from the incentive architecture of intermediated capital deployment, rather than from a failure of implementation.

The gap is not permanent. Five months later, when MAS named the first three EQDP asset managers (Avanda Investment Management, Fullerton Fund Management, and JP Morgan Asset Management) with mandates requiring "significant allocation to small- and mid-cap stocks," the cross-sectional pattern reversed. The average equity gained +3.1 percent, with over 70 percent of firms posting positive abnormal returns, and the size gradient flipped sign: smaller firms gained more than larger ones. The intermediation gap appeared to narrow once the market received

[1] The removal of public oversight effectively revitalises the market for lemons (Akerlof, 1970).

credible information about mandate design.

Neither instrument's failure was uniform. The capital evidence shows that the intermediation gap is design-sensitive: once credible mandate terms were disclosed, the size gradient reversed. The information channel presents an analogous nuance. For most high-friction firms, the market penalised the proposed removal of oversight because it would lose a disciplinary signal. But for financially distressed firms trapped in a self-reinforcing cycle of watch-list inclusion, financing impairment, and low valuation, the regulatory machinery itself was part of the problem. The friction count measures accumulated scrutiny, which the market reads as informative monitoring. The watch-list, by contrast, is a specific institutional mechanism that imposes concrete costs on affected firms, some of which resorted to dilutive share issuances solely to satisfy the S$40 million market capitalisation threshold. Conditional on overall friction counts, watch-list status predicts a positive +5.2 percent abnormal return under FF3 (+3.2 percent under CAPM) at E1. The FF3 coefficient is larger because the SMB factor absorbs general size variation, leaving the watch-list indicator to capture the pure stigma-relief premium net of the market capitalisation threshold that defines watch-list eligibility. The market priced watch-list removal as a discrete benefit for the firms most directly affected by the stigma mechanism. When the watch-list was formally abolished on October 29, 2025, the five previously watch-listed firms in our sample gained +12.7 percent on average under FF3, compared to +1.0 percent for non-watch-listed firms. The small treated sample limits statistical precision, but the direction is consistent with deregulation creating value when it removes an identified trap rather than dismantling oversight wholesale.

The intermediation gap and the friction penalty are thus two instances of a single phenomenon: the distance between regulatory intent and market outcome, mediated by the institutional structure through which policy is transmitted. Capital channelled through professional intermediaries flows where intermediary incentives direct it, not where policy directs it. Deregulation channelled as a blanket reduction in oversight removes valuable information alongside genuine burden. Both failures are predictable from institutional architecture, and both are design-sensitive: specific mandate terms narrowed the intermediation gap, and targeted stigma removal created value where wholesale oversight removal destroyed it. We refer to the two pricing channels as *intermediation* (the market's anticipation that fund manager constraints would redirect capital toward large firms, proxied empirically by firm size) and *information* (the market's interpretation of deregulation as removing informative monitoring, proxied by accumulated regulatory friction).

The paper makes three contributions. First, we document the regulatory intermediation gap empirically and show that it operates in public equity markets, extending the directed lending and government procurement literatures to a new institutional context. The gap is not permanent: when credible mandate terms are disclosed, the market revises, providing evidence that the design of intermediated capital deployment matters as much as its quantum.

Second, we demonstrate that a single bundled reform announcement is simultaneously priced on two independent dimensions—intermediation, captured by size, and information loss, captured by friction—providing a within-event methodology applicable to the bundled policy interventions common in securities regulation. Both dimensions reflect the same underlying problem: policy transmitted through institutional channels is shaped by those channels' own constraints.

Third, we show that different components of regulatory friction carry different prices, offering a more nuanced view of deregulation than the standard "regulation as cost" framework:[2] the market penalises the removal of oversight that provides informative monitoring, but rewards the removal of an identified stigma mechanism (the financial watch-list) whose institutional costs exceed its informational value. Decomposing friction by type reveals that halt and suspension events drive the entire penalty while routine queries are not priced. Decomposing by recency shows that accumulated historical friction drives the effect while recent friction does not, indicating that the market treats regulatory attention as a stock of reputation rather than a flow of current scrutiny.

The remainder of the paper is organised as follows. Section 2 situates the paper in the relevant literatures. Section 3 describes the institutional background of Singapore's equity market and the reform timeline. Section 4 describes the data and methodology. Section 5 develops the empirical specifications and presents the results. Section 6 discusses limitations, and Section 7 concludes.

2 Literature Review

This paper sits at the intersection of three literatures: the efficacy of government-directed financial interventions, the informational value of securities regulation, and the institutional dynamics of Singapore's equity market. We review each in turn, identifying the gap that motivates our empirical design.

2.1 Intermediated Government Interventions and the Small-Cap Capital Deficit

Private intermediaries systematically redirect government-directed capital away from its intended beneficiaries. Banerjee and Duflo (2014) show that expanding eligibility for directed lending relaxed binding credit constraints for newly eligible firms, confirming that intermediated capital is rationed in ways that exclude targeted recipients. The same dynamic appears in government procurement, where affirmative action provisions are diluted by intermediary contractor behaviour (Marion, 2009). In monetary policy, Krishnamurthy and Vissing-Jorgensen (2011) demonstrate that quantitative easing operates through segmented markets in which portfolio balance effects concentrate in the largest, most liquid asset classes; subsequent work confirms that the scale and composition of purchases matter for transmission (Gagnon et al., 2011; D'Amico and King, 2013). At the policy level, the Kay (2012) diagnosed how the proliferation of intermediation in equity investment had led to misaligned incentives and a tendency to evaluate market effectiveness through the lens of intermediaries rather than companies or end investors. Earlier UK reviews reached complementary conclusions: the Myners (2001) on misaligned incentives in the institutional investment chain and the Walker (2009) on governance failures in financial institutions. The phenomenon that Braun and Christophers (2024) label "asset manager capitalism"—the growing

[2] Lawrence (2000) frames this as the tension between the costs of market confidence and the costs of regulation.

dominance of intermediary constraints over end-investor interests—is precisely the mechanism our paper measures empirically. Narrow mandates and agency frictions in delegated management limit deviations from benchmarks, providing theoretical foundations for this pattern (He and Xiong, 2008; Buffa et al., 2014).

These intermediation frictions operate against a backdrop of structural decline in small-cap public equity markets. The number of publicly listed firms has fallen sharply across developed markets (Doidge et al., 2017; Kahle and Stulz, 2017), driven in part by the collapse of small-company IPO activity (Gao et al., 2013). Grullon et al. (2019) document a related trend: increasing industry concentration that coincides with a growing share of institutional capital flowing to the largest firms. The growth of index-linked investing has reshaped price formation and liquidity allocation, with adverse consequences for non-index securities (Wurgler, 2010; Israeli et al., 2017), while institutional investors systematically tilt toward large-cap, benchmark-proximate holdings (Bennett et al., 2003), and fund managers who deviate least from their benchmarks—closet indexers with low active share—mechanically concentrate capital in index-heavy names (Cremers and Petajisto, 2009). This institutional tilt is rational given a market structure in which capital concentration and return concentration reinforce each other.

Government programmes designed to counteract these trends span distinct categories: some route capital through institutional investors (Japan's Government Pension Investment Fund allocation shifts), others are broader market-structure or listing reforms (Korea's Corporate Value-Up Programme, the European Union's Listing Act of December 2024 (European Parliament and Council of the European Union, 2024)), and still others operate through bank-capital rules (the EU's SME Supporting Factor under Article 501 of the Capital Requirements Regulation, which the European Banking Authority (2016) found had limited effect on SME credit supply). Despite this proliferation, the intermediation problem in equity markets has received almost no empirical attention. The fund managers through which public market capital must flow are themselves optimised for large, liquid, benchmark-proximate securities. Crucially, the existing literature documents intermediation gaps as static phenomena—capital is mis-routed or it is not—without testing whether the gap responds to the specificity of mandate design. Our contribution is twofold: we show that the market priced a likely intermediation gap in a public equity setting, and we exploit cross-event variation in mandate specificity (the E1-to-E3 reversal) to demonstrate that the gap is design-sensitive, observable in announcement returns rather than inferred from ex post credit allocation data.

2.2 The Informational Value of Securities Regulation

Our information channel connects to a longstanding debate about whether securities regulation creates or destroys value. A foundational literature establishes that legal institutions and investor protection shape financial development (La Porta et al., 1997, 1998; Shleifer and Wolfenzon, 2002), and that the design of securities laws matters for market outcomes (La Porta et al., 2006). One strand of this literature treats regulation primarily as a cost, predicting that deregulation should be welcomed by the market. Stigler (1971) and Peltzman (1976) provide the theoretical foundations;

applied to securities regulation, event studies of the Sarbanes–Oxley Act document significant compliance costs (Zhang, 2007; Iliev, 2010), and the JOBS Act generated positive returns for firms exempted from disclosure requirements (Dharmapala and Khanna, 2014). Yet commitment to stricter oversight can itself lower the cost of capital by reducing adverse selection (Carpentier et al., 2010).

Our finding that the market penalised high-friction firms at a deregulation announcement supports an alternative view: securities regulation serves an informative monitoring function whose removal increases information asymmetry (Healy and Palepu, 2001; Leuz and Verrecchia, 2000; Christensen et al., 2016). Disclosure mandates generate positive abnormal returns and liquidity improvements (Bushee and Leuz, 2004; Greenstone et al., 2005; Ferrell, 2007), while deregulation episodes produce significant negative market reactions (Fernandes et al., 2008; Jenkinson and Ramadorai, 2008; Hoffman et al., 2018; Hitz and Moritz, 2019; Behrmann et al., 2024). Enforcement intensity matters as much as formal standards: higher enforcement is associated with a lower cost of capital (Coffee, 2007; Christensen et al., 2013). These competing regulatory philosophies have deep roots. When Singapore shifted towards disclosure-based regulation, the approach was characterised as "caveat emptor";[3] by contrast, the original US disclosure regime was framed as ensuring that "the seller also beware."[4] Our friction results suggest the market's assessment lies closer to the latter view: removing oversight shifts risk to buyers. In a related vein, Nielsson (2013) finds that London's AIM, which imposes lighter listing requirements, allows small firms to raise more capital without attracting observably lower-quality issuers; the sorting operates through firm size and sensitivity to regulatory costs, a dimension directly relevant to the Singapore context where smaller, higher-friction firms bore the deregulation penalty.

The adverse selection dimension of our result connects to the market-for-lemons logic (Akerlof, 1970): mandatory disclosure prevents market breakdown by enabling investors to distinguish between clean and problematic firms (Coffee, 1984). Our finding that the market distinguishes between general friction (penalised) and watch-list status (rewarded) resonates with work on regulatory stigma, where the regulatory mechanism itself compounds the problem it was designed to address. Correia (2014) documents how political connections shape enforcement outcomes, suggesting that regulatory actions carry heterogeneous information content depending on firm characteristics. Reputational damage from regulatory sanctions can dwarf direct financial penalties (Armour et al., 2010, 2019), and enforcement actions generate positive valuation spillovers for non-target firms through the legal bonding channel (Silvers, 2016). In banking, stigma from institutional labelling creates analogous feedback loops: banks avoided the Federal Reserve's discount window during the 2007–2008 crisis because borrowing signalled distress (Armantier et al., 2015; Anbil, 2018). The watch-list mechanism we study operates through a similar channel: the label imposed costs—financing impairment, counterparty reluctance—that exceeded its informational content, so its removal was priced as relief rather than information loss. Financial distress compounds these stigma effects: highly leveraged firms lose market share during downturns (Opler and Titman, 1994) and face transaction costs that trap them in suboptimal capital structures (Gilson, 1997), suggesting that the watch-list's impact on affected firms extended through both informational

[3] Lee Hsien Loong, Minister for Trade and Industry, speech at the 1st Anniversary Dinner of the Securities Investors Association (Singapore), 7 July 2000.

[4] Franklin D. Roosevelt, Message to Congress on Federal Supervision of Investment Securities, 29 March 1933.

and competitive channels.

Our event study design draws on well-established methods for using security prices to assess regulatory announcements (Schwert, 1981; MacKinlay, 1997; Khoo et al., 2025). Because our design studies common announcement dates rather than staggered treatment adoption, it draws on the classic cross-sectional financial event study rather than the panel event-study framework of Schmidheiny and Siegloch (2019), though that work informs our approach to multi-event analysis. Broader work on securities-law design and enforcement includes Mahoney (1997) and Jackson and Roe (2009).

2.3 Singapore's Equity Market and Corporate Governance

Singapore's equity market provides a distinctive institutional setting for testing both channels. The development of the city-state's corporate law has historically been highly responsive to economic events, creating a uniquely reciprocal relationship between regulatory architecture and market outcomes (Ooi and Tan, 2019). This responsiveness means that reform announcements carry credible information about near-term regulatory change, strengthening the event-study design. Lee (2013) provides a detailed microstructure analysis of SGX trading processes, while Claessens et al. (2006) situate small-exchange development within the broader dynamics of stock market internationalisation.

Research on corporate governance in Singapore centres on concentrated ownership, family control, and government-linked companies (Tan et al., 2014). State ownership and concentrated blockholders interact to shape governance outcomes and minority protections in ways central to understanding the market's institutional constraints (Lim, 2021). The determinants of corporate ownership and board structure reflect the influence of concentrated shareholdings on governance practices (Mak and Li, 2001), a theme explored further in studies of independent directors and board composition (Lim, 2019; Puchniak and Lan, 2017). These ownership structures matter for our analysis because they imply that investability constraints bind more tightly in Singapore than in markets with dispersed ownership: a concentrated shareholder base and thin float reinforce the intermediation gap by limiting the free-float available to institutional investors.

The regulatory architecture we study—specifically SGX's public query regime, the financial watch-list, and the post-listing oversight framework—has features particular to this institutional context. The watch-list mechanism has no direct analogue in major comparator markets and represents an unusually visible form of regulatory labelling (Tjio et al., 2017).[5] The absence of a comparator mechanism makes the watch-list abolition a particularly clean test of how the market values regulatory labelling: there is no substitute oversight regime that could offset the information loss.

The closest antecedent to our work is Gurrea-Martínez (2025), who provides a comprehensive qualitative assessment of the reform's potential effects, situating Singapore's package within a

[5] Wan et al. (2017) compare public and private enforcement in Hong Kong and Singapore, documenting the dominance of public enforcement, and Yi (2024) analyse SGX's alternative board design in comparative perspective with Hong Kong and Malaysia.

comparative framework of international market competitiveness initiatives. Our paper complements that analysis by testing empirically whether the market priced the reform's components as the policy intended. We contribute to this Singapore-specific literature by providing the first event study of SGX's 2025 reform package, documenting how the market's concentrated size distribution and steep investability threshold interact with intermediated capital deployment in ways that may not arise in deeper, more diversified equity markets. The dual-listing bridge raises a related selection question: Iliev et al. (2013) find that loosening US regulations attracted involuntary cross-listings with mixed valuation consequences, cautioning that regulatory bridges may have unintended selection effects.

3 Institutional Background

3.1 Singapore's Equity Market and Its Challenges

SGX occupies an unusual position in the global exchange ecosystem: a well-regulated, highly credible venue that nevertheless struggles to attract new listings and sustain trading interest in existing ones. Over the past decade, market participants and policymakers have raised persistent concerns about declining IPO activity, thinning trading volumes, and stagnating market capitalisation relative to regional competitors in Hong Kong, Tokyo, and Sydney. Small- and mid-cap equities bear the brunt of these trends. Thin analyst coverage reduces investor awareness, low institutional participation depresses valuations, wide bid-ask spreads discourage trading, and the resulting poor market quality deters new companies from listing. This self-reinforcing decline ultimately prompted the government intervention examined in this paper.

The exchange operates two listing boards. The Mainboard hosts larger, more established companies and imposes quantitative admission criteria including a minimum pre-tax profit threshold (S$30 million, subsequently reduced to S$10 million in October 2025) and market capitalisation requirements. Catalist, established in 2007, provides a more flexible sponsor-supervised regime for smaller growth companies. In practice, however, Catalist has attracted relatively few high-quality listings, and many of its constituent firms trade infrequently at deep valuation discounts.

3.2 The Watch-List and Post-Listing Regulatory Apparatus

A distinctive feature of SGX's regulatory architecture is the financial watch-list, introduced in 2008 as a publicly maintained register of Mainboard firms that had recorded pre-tax losses for three or more consecutive financial years and whose average daily market capitalisation fell below S$40 million over the preceding six months.[6] Inclusion carried substantial stigma. Market participants, lenders, and business counterparties commonly treated watch-list status as a signal

[6] There was no equivalent watch-list for SESDAQ (later Catalist) companies, which were expected to be monitored by their sponsors.

of impending delisting risk, further impairing the affected firm's ability to secure financing, attract investors, or restructure. Within two years of the watch-list's introduction, five of its six inaugural constituents had been delisted.[7]

The watch-list's composition changed markedly over the following decade. At its peak in 2016, approximately 75 companies appeared on the list. Following the formation of SGX RegCo in 2017, the count declined through delistings, suspensions, and exits meeting the financial criteria. Issuers listed solely because of the minimum trading price criterion gained an additional exit route when that requirement was removed for Catalist transfers in June 2020. By mid-2025, roughly 30 companies remained.[8] The watch-list was formally abolished in October 2025 as part of SGX's broader shift towards a disclosure-based regime. SGX RegCo justified the removal by citing the watch-list's unintended negative effects on business confidence and access to financing. The firms remaining on the list by 2025 were also survivors of a protracted filtering process, which may have reduced the expected cost of abolition relative to an earlier removal.

Beyond the watch-list, SGX RegCo maintains a broader apparatus of post-listing regulatory interventions: public queries (issued when unusual trading activity is detected), trade-with-caution alerts, and trading suspensions for firms with going-concern issues. These interventions are individually disclosed. Together with watch-list status, they form a cumulative record of regulatory attention that underpins the friction measure used in the empirical analysis.

3.3 The Equities Market Review Group

On August 2, 2024, the Monetary Authority of Singapore (MAS) announced the formation of the Equities Market Review Group. Chaired by Mr Chee Hong Tat—Minister for Transport, Second Minister for Finance, and Deputy Chairman of MAS—the panel brought together senior private and public sector representatives. The Review Group's mandate was to recommend measures to strengthen the competitiveness of Singapore's equities market, with a twelve-month target for completion.

Two workstreams supported the Review Group, each corresponding to one of the reform's principal instruments. The Enterprise and Markets workstream focused on fostering listings, increasing investor participation, improving trading liquidity, and facilitating fair valuation. These objectives would be pursued primarily through *capital deployment*, including the S$5 billion Equity Market Development Programme (EQDP) channelled through professional fund managers. The Regulatory workstream focused on streamlining the regulatory framework, improving the listing process, enhancing the disclosure-based regime, and strengthening corporate governance standards and investor recourse. These objectives would be pursued through *deregulation* of SGX's post-listing oversight apparatus. This dual-instrument structure—capital deployment on one track, deregulation on the other—is central to the empirical design because the reform events progressively

[7] https://sbr.com.sg/financial-services/news/singapore-exchange-delist-5-companies-under-watch-list

[8] Our equity-only regression sample contains five firms with active watch-list status at the October 2025 removal event. The difference from the full watch-list reflects firms that were suspended from trading, delisted prior to the event window, or excluded from our sample as non-equity instruments.

unbundle the two instruments.

3.4 Timeline of Reform Events

The reform process unfolded through a series of discrete announcements over sixteen months (Table 1). This sequence is central to the empirical design: the first substantive announcement (E1) bundled both instruments in a single event, while subsequent events activated narrower, though partially overlapping, subsets of these channels, creating the cross-event variation in policy content that structures the analysis.

3.4.0.1 Signal event.

On August 2, 2024, MAS announced the Review Group's formation (E0), a signal event that established the policy direction but contained no specific measures.

3.4.0.2 First measures.

The first substantive announcement came on February 21, 2025 (E1), when the Review Group published its first set of measures across six pillars: supply (tax incentives, Catalist review, financing support), demand (S$5 billion EQDP, fund manager tax exemptions, GIP Option C adjustment, GEMS research grant expansion), cross-border partnerships (depository receipt collaborations), a decisive shift towards a disclosure-based regime (consolidation of listing review under SGX RegCo, streamlined qualitative admission criteria, streamlined prospectus requirements), strengthened investor confidence (investor recourse avenues, shareholder engagement), and improved trading and settlement efficiency.

3.4.0.3 Progressive unbundling.

Subsequent events progressively unbundled the February package along two parallel tracks. On the deregulation side, SGX RegCo published its consultation paper on a more disclosure-based regime at E2 (May 15, 2025), proposing the removal of the financial watch-list, a reduction of the Mainboard profit threshold, streamlined qualitative admission criteria, a shift from public queries to private engagement, time-limited trade-with-caution alerts, and narrowed trading suspension criteria. On October 29, 2025 (E4), SGX RegCo implemented these changes with immediate effect, removing the financial watch-list and ceasing public trading queries.

On the capital-deployment side, MAS appointed the first three EQDP asset managers—Avanda Investment Management, Fullerton Fund Management, and J.P. Morgan Asset Management—at E3 (July 21, 2025). It deployed S$1.1 billion in initial capital, disclosed mandate terms requiring allocation to small- and mid-cap equities, committed S$50 million to equity research and the

listed product ecosystem, and outlined proposals to strengthen investor recourse. This disclosure provided the market with concrete information about how the capital would be deployed. On November 19, 2025 (E5), the Review Group published its final report alongside the SGX-Nasdaq dual listing bridge (allowing SGX-listed firms to obtain a secondary Nasdaq listing through depositary receipts with reduced regulatory friction), a second batch of six EQDP managers with S$2.85 billion, and market structure enhancements directly targeting the liquidity deficit: a board lot reduction[9] and designated market maker incentives.[10] These liquidity measures are consistent with the scale economies documented by Kervel et al. (2025) in the context of Chile's tax reform. The sequential unfolding of these events—each activating a different subset of capital deployment and deregulation measures—provides the cross-event variation that structures the empirical analysis.

Table 1: Timeline of Reform Events

Label	Date	Key Content	Primary Channels
E0	2 Aug 2024	Formation of Equities Market Review Group	Signal (no measures)
E1	21 Feb 2025	First measures: S$5B EQDP, disclosure-based regime shift, watch-list removal proposal, Catalist review, prospectus streamlining	Capital + Deregulation
E2	15 May 2025	SGX RegCo consultation paper: detailed disclosure-based proposals	Deregulation
E3	21 Jul 2025	First 3 EQDP managers named (Avanda, Fullerton, JPMAM); S$1.1B deployed; small/mid-cap mandate terms disclosed	Capital
E4	29 Oct 2025	Watch-list abolished; public trading queries ceased	Deregulation (implementation)
E5	19 Nov 2025	Final report; SGX–Nasdaq dual listing bridge; 6 additional EQDP managers (S$2.85B); board lot reduction, market maker incentives	Capital + Market structure

Notes: "Primary Channels" indicates which of the paper's two main channels, namely capital deployment (intermediation) and deregulation (information), each event primarily activates. E1 bundles both; subsequent events progressively unbundle them.

[9]SGX moved from a minimum trading unit of 100 shares to a single share, lowering minimum transaction sizes and reducing the effective cost of trading for illiquid securities.

[10]Fee rebates for firms committing to continuously quote bid and ask prices, guaranteeing tradability for thinly traded securities.

4 Data and Methodology

4.1 Data Sources

The analysis combines SGX price data with firm-level characteristics in an equity-only event-study design. Daily end-of-day prices for all SGX-listed securities come from SGX's data services division, covering August 1, 2022 to August 29, 2025 (1,048,575 firm-day observations across 3,112 stock codes). For events after August 2025, Yahoo Finance daily close and volume data extend the price series through December 31, 2025 (60,865 observations for 716 stock codes). A 20-day overlap window validates the Yahoo data against the original SGX series (median price correlation: 1.000).[11]

Factor model inputs consist of Fama–French three-factor daily returns (Fama and French, 1993) for the primary CAR specification and five-factor daily returns (Fama and French, 2015) for robustness checks, both for the Asia Pacific ex Japan region and obtained from the Dartmouth Data Library. Regional rather than global factors follow Fama and French (2012), who show that a regional model captures size, value, and momentum premia in international stock returns more effectively than a global specification. The SGX equal-weighted market return, computed daily from all listed equities with valid price observations, serves as the market benchmark for the market model (CAPM), which enters as a secondary specification.

Firm characteristics draw on multiple sources. Market capitalisation comes from SGX monthly listings snapshots (705 stock codes, 37 months). Annual financial statement data originate primarily from Worldscope via WRDS, supplemented by firm-level financial statements extracted from approximately 17,000 SGX filing PDFs using a three-pass extraction pipeline.[12] Regulatory friction measures aggregate 31 months of SGX corporate announcement CSVs (≈100,540 announcements), parsed into eight friction categories. Board composition data rely on SGX's XML issuer feeds and the Company with Active Directors registry (≈3,926 director-company links); sector classifications use SSIC codes matched from the same feeds.

4.2 Sample Construction

The analysis sample retains only equity securities, excluding warrants, structured products, daily leverage certificates, exchange-traded funds, REITs, and business trusts using classification flags in SGX's XML security feed.[13] The resulting firm-event panel yields approximately 260–275 firms

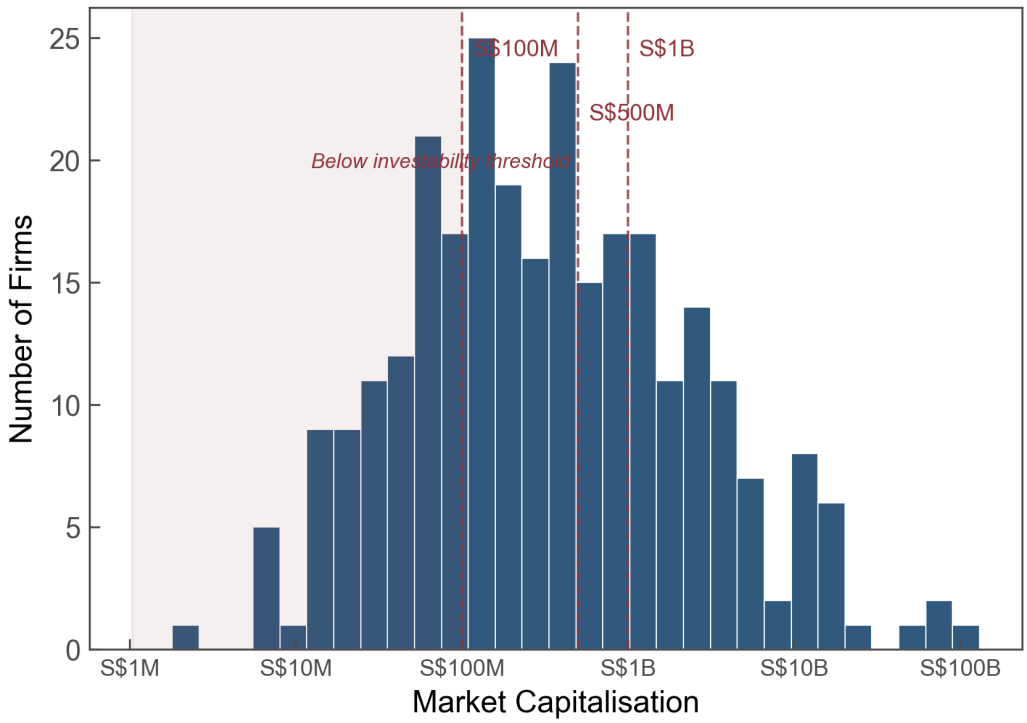
[11] Yahoo Finance data do not include bid-ask quotes, so quoted spread and exact traded value are unavailable for the extended period. This is immaterial for CAR estimation but precludes post-event market quality analysis for the E4 and E5 events.

[12] The pipeline classified 204,727 files across the 97.5 GB SGX filing corpus: pdfplumber for structured tables (≈70% success rate), Qwen2.5-VL vision model via local Ollama inference for unstructured layouts, and manual review flags for ≈70 rows with known scale or currency ambiguities.

[13] Excluding non-equity instruments materially affects certain results. The Catalist board-tier effect reported in preliminary analyses (−3.0 percentage points, $p < 0.001$) was substantially inflated by structured products, exclusively

per event with valid CARs and cross-sectional data: 260 at E0, 274 at E1, 275 at E2, 270 at E3, and 258 at both E4 and E5. Because the Asia Pacific ex Japan factor file provides fewer overlapping trading days with SGX-listed securities than a US factor file, some firms fall below the minimum 120-observation threshold imposed on the estimation window. The sample is therefore slightly smaller than the full set of equity stock codes but provides consistent factor model coverage across all events.

Figure 1: Distribution of market capitalisation across the SGX equity-only sample, shown on a log-scaled x-axis. Dashed vertical lines mark illustrative investability thresholds at S$100 million, S$500 million, and S$1 billion; the shaded region highlights firms below the S$100 million threshold.



Note: The shaded region highlights firms below the S$100 million threshold. Market capitalisation is from the last available SGX monthly snapshot before E1.

The pronounced right skew visible in Figure 1 motivates the median split used in Table 5 and the quartile analysis in Table 6: a substantial share of SGX-listed equities falls below the investability thresholds that professional fund managers typically impose.

Table 2 reports summary statistics for the E1 cross-section. Of the 274 firms with valid CARs, broker ownership data are available for approximately 205 and accounting variables (ROA, leverage) for approximately 257. Directly treated groups are rare—watch-list firms comprise 1.8% of the sample

Mainboard-listed, whose large positive CARs depressed the Mainboard reference group mean. After equity-only filtering, the Catalist effect attenuates to −1.8 percentage points ($p = 0.136$) and is absorbed by the size control.

($N = 5$) and distressed firms 8.8%—constraining precision in the corresponding cross-sectional tests.

Table 2: Descriptive Statistics (E1 Sample)

	N	Mean	SD	Min	Median	Max
CAR$(-1, +3)$	274	-0.013	0.055	-0.195	-0.006	0.115
log(Market cap)	283	19.637	1.958	15.827	19.501	24.774
Friction count	285	2.475	3.302	0.000	1.000	17.160
Watchlist	285	0.018	0.132	0.000	0.000	1.000
Above median	285	0.498	0.501	0.000	0.000	1.000
ROA (%)	257	1.848	14.242	-78.143	3.855	19.459
Leverage	258	23.874	18.016	0.000	22.869	67.866
Book-to-market	249	1.540	1.111	0.032	1.325	5.034
Illiquidity (z-Amihud)	283	-0.176	0.159	-0.217	-0.217	0.749
Catalist	285	0.147	0.355	0.000	0.000	1.000

Notes: Sample comprises 285 SGX-listed equities in the E1 cross-section (2025-02-21); 274 have valid FF3 CARs. Market cap is from the last available trading day before the event window. Illiquidity (z-Amihud) is the standardised Amihud (2002) measure over the 60 trading days before the event. Friction count is the cumulative number of SGX queries, trading halts, and trading suspensions accumulated by the firm as of the event date; watch-list and distress are reported separately as indicator variables. Broker ownership is the share of top-20 holdings registered through retail securities brokers; available for 205 firms with shareholder register data. Accounting variables (ROA, Leverage) are available for firms with Worldscope coverage ($N = 257$–258). Catalist, watch-list, and distress are indicator variables.

4.3 Variable Definitions

The cross-sectional analysis relies on two primary explanatory variables, each corresponding to one of the paper's two pricing channels. The *intermediation channel* is captured by firm size. The primary specification uses an indicator variable, *above_median*, equal to one if the firm's log market capitalisation exceeds the event-specific sample median. Continuous log market capitalisation serves as a secondary robustness measure. In both cases, market capitalisation is measured from the last available SGX monthly snapshot before the event window.

The *information channel* is captured by a regulatory friction count: the cumulative number of SGX queries, trading halts, and trading suspensions that a firm has accumulated as of the event date, constructed from the parsed announcement data described above. Two refinements probe the friction measure further. First, friction events are decomposed by type—queries versus halts and suspensions—to distinguish routine monitoring from consequential regulatory action. Second, the count is split by recency—events within the 12 months preceding the announcement versus earlier events—to separate current scrutiny from accumulated reputation.

A *watch-list* indicator equals one for the five firms on SGX's Financial Watch-List at the time of

E4 abolition.[14] A *distress* indicator equals one for firms meeting SGX's financial distress criteria. Illiquidity is the standardised Amihud (2002) measure, computed over the 60 trading days preceding the event window and denoted z-Amihud. The Catalist indicator equals one for firms listed on the Catalist sponsor-supervised board.

Financial controls—return on assets (ROA), leverage, and book-to-market ratio—are drawn from the most recent annual financial statements. All cross-sectional variables are measured before the event window and, for post-E1 events, forward-filled from the pre-E1 snapshot to avoid conditioning on post-treatment realisations. Sector fixed effects use two-digit SSIC codes.

4.4 CAR Estimation

A key design choice is that all post-E1 events share a fixed pre-E1 estimation window, ensuring that factor loadings are not contaminated by the reform itself. Cumulative abnormal returns are estimated event by event using the Maynes and Rumsey (1993) thin-trading-adjusted procedure. The method regresses firm returns on contemporaneous and lagged factor returns in a single OLS pass without an intercept (no-constant specification), then cumulates the combined residuals over the event window. This adjustment addresses the non-synchronous trading bias that is prevalent in thin markets such as SGX, where many securities do not trade every day. Firms with fewer than 120 valid return observations in the estimation window are excluded to ensure reliable factor loading estimates.

The market model regresses firm i's daily return on the SGX equal-weighted market return:

$$R_{i,t} = \hat{\beta}_i R_{m,t}^{\mathrm{EW}} + \hat{\varepsilon}_{i,t} \tag{1}$$

and cumulates the residuals $\hat{\varepsilon}_{i,t}$ over the event window. The three-factor model replaces the single market factor with the Asia Pacific ex Japan MKT, SMB, and HML factors:

$$R_{i,t} = \hat{\beta}_{1,i}\mathrm{MKT}_t + \hat{\beta}_{2,i}\mathrm{SMB}_t + \hat{\beta}_{3,i}\mathrm{HML}_t + \hat{\varepsilon}_{i,t} \tag{2}$$

Under the no-constant specification, the intercept is suppressed in both models; the abnormal return is the full residual rather than the deviation from an estimated alpha.[15] The Fama–French three-factor model using Asia Pacific ex Japan factors serves as the primary specification throughout; the market model enters as a secondary specification in three specific contexts where the SMB factor mechanically absorbs variation of interest.

The primary event window spans $(-1, +3)$ trading days around the event date; robustness checks use $(-1, +1)$, $(-1, +5)$, $(0, +2)$, and $(0, +5)$.

[14] The five watch-listed firms at E4 are A31, AZA, FQ7, QS9, and R14. A sixth firm, 5GI, had been delisted before the event date.

[15] The no-constant approach is standard in thin-trading adjustments following Maynes and Rumsey (1993). Including an intercept in a thin market can absorb genuine abnormal returns into the alpha estimate when many zero-return days are present.

The mean cumulative abnormal return for a group G of N_G firms is:

$$\overline{\text{CAR}}_G = \frac{1}{N_G} \sum_{i \in G} \text{CAR}_{i,e} \tag{3}$$

with statistical significance assessed by a two-sided t-test against zero. When firms are sorted into size quartiles or above and below the median of log market capitalisation, G indexes the relevant partition.

For E0, the estimation window spans the standard $[-250, -30]$ trading days before the event. For post-E1 events (E2, E3, E4, E5), the fixed pre-E1 window runs from January 2, 2024 to February 14, 2025 ($\approx$280 trading days), avoiding contamination from the E1 structural break.[16]

4.5 Cross-Sectional Regression Framework

The second stage regresses firm-level CARs on cross-sectional characteristics using ordinary least squares with HC1 heteroskedasticity-robust standard errors. Stacked specifications that pool multiple events include event fixed effects and cluster standard errors by stock code. All CARs are winsorised at the 1st and 99th percentiles.

To trace how each pricing channel operates, the cross-sectional regressions follow a stepwise five-specification progression that is common to all main regression tables. Specification (1) enters only the variable of interest—*above_median* for the intermediation channel or *friction* for the information channel—to establish the unconditional association. Specification (2) adds the other channel's variable, testing whether the two channels absorb one another. Specification (3) introduces financial controls (ROA, leverage, and book-to-market), which are reported as a single "Financial controls" indicator row rather than as individual coefficient rows, since they serve as nuisance controls rather than quantities of interest. Specification (4) replaces the financial controls with illiquidity (z-Amihud) and the Catalist board indicator, probing whether the channel of interest survives conditioning on market microstructure and listing-tier effects. Specification (5) is fully saturated, combining all controls with two-digit SSIC sector fixed effects.

This progression serves two purposes. First, it allows the reader to assess robustness by observing whether the coefficient of interest remains stable across increasingly demanding specifications. Second, because the two channel variables enter together from Specification (2) onward, it directly tests whether intermediation and information are genuinely independent pricing dimensions.

Because the second-stage regressions use estimated dependent variables, standard inference may overstate precision. To address this concern, a paired-date reversal placebo test is conducted (see Section A.1.1). Two hundred pairs of pseudo-event dates are sampled uniformly from the pre-E1 period (June 2023 to January 2025), with each pair separated by approximately 105 trading days

[16] The gap between the estimation window endpoint and the E2 event date is approximately 60 trading days. If E1 permanently altered stock betas, the pre-E1 estimates may be stale. A robustness check using inter-event estimation windows produces qualitatively similar cross-sectional coefficients with wider standard errors.

to match the E1–E3 gap. For each pair, CARs are estimated at both dates and the cross-sectional coefficients computed; the reversal statistic $\Delta\hat{\beta} = \hat{\beta}_{\text{date}_2} - \hat{\beta}_{\text{date}_1}$ provides a non-parametric null for whether the observed E1-to-E3 sign flips in the size and friction coefficients could arise by chance. The Results section that follows applies this framework, with each subsection developing the motivating economic logic, stating the specification, and presenting the estimates for one channel or interaction at a time.

5 Results

This section reports the empirical results. Each subsection develops the motivating logic, states the specification, and presents the estimates.

5.1 Event-Level Summary

Table 3 reports mean CARs (Equation (3)) across the six events in the reform timeline. The first substantive announcement (E1, February 21, 2025) produced a negative mean CAR of −1.32% ($t = -3.95$, $p < 0.001$, $N = 274$), with only 40.5% of firms posting positive returns: the typical firm lost value on the announcement. The consultation event (E2, May 15, 2025) elicited a null response (−0.11%, $p = 0.721$), consistent with its content having been fully anticipated from E1. By contrast, the EQDP manager appointment (E3, July 21, 2025) produced a strongly positive mean CAR of +3.12% ($t = 6.82$, $p < 0.001$, $N = 270$), with 70.0% of firms posting positive returns; the actual capital commitment was uniformly positive. The watch-list removal (E4) generated a positive mean CAR of +1.20% ($p = 0.003$), and the final report (E5) was also positive (+0.62%, $p = 0.006$). The contrast between the negative E1 and the strongly positive E3 motivates the cross-sectional analysis that follows.

Table 3: Event-Level Summary Statistics

Event	Date	Channel	N	Mean CAR	t-stat	p-value	Median	% Pos
E0	2024-08-02	Signal	260	-0.0215^{***}	-5.25	0.000	-0.0187	30.8%
E1	2025-02-21	Capital + Deregulation	274	-0.0132^{***}	-3.95	0.000	-0.0058	40.5%
E2	2025-05-15	Deregulation	275	−0.0011	-0.36	0.721	-0.0037	42.9%
E3	2025-07-21	Capital	270	0.0312^{***}	6.82	0.000	0.0199	70.0%
E4	2025-10-29	Deregulation (impl.)	258	0.0120^{***}	2.99	0.003	0.0035	55.8%
E5	2025-11-19	Capital + Market structure	258	0.0062^{***}	2.76	0.006	0.0063	60.5%

Notes: The dependent variable is the firm-level cumulative abnormal return over the $(-1, +3)$ trading-day window, estimated using a Fama–French three-factor model with Asia Pacific ex Japan factors. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ for H_0: mean = 0.

5.2 The Intermediation Gap

Government-intermediated equity market interventions are systematically redirected away from their intended beneficiaries when channelled through professional fund managers. The EQDP deploys S$5 billion through managers subject to standard institutional constraints: fiduciary duties that favour established firms on a risk-adjusted basis,[17] AUM-driven fee structures that penalise small positions, and liquidity screens that exclude thinly traded securities from the investable universe. If the market correctly anticipates these constraints, the announcement should produce a cross-sectional pattern in which larger firms benefit disproportionately, regardless of the stated policy objective of supporting small- and mid-cap equities. (Liquidity is examined separately as a conditional screen within the investable set; see Section 5.5.)

The dependent variable throughout is $\text{CAR}_{i,e}$, the cumulative abnormal return (CAR) for firm i around event e, estimated using a Fama–French three-factor model (Fama and French, 1993) with Asia Pacific ex Japan factors over a $(-1, +3)$ trading day event window. The baseline specification is:

$$\text{CAR}_{i,\text{E1}} = \alpha + \beta_1 \text{AboveMedian}_i + \gamma' X_i + \varepsilon_i \tag{4}$$

where AboveMedian_i equals one if firm i's market capitalisation exceeds the within-event-sample median, measured from the most recent monthly listings snapshot prior to the event date, and the control vector X_i includes return on assets, leverage (total liabilities over total assets), and book-to-market ratio, all from the most recent fiscal year. Standard errors are HC1 robust and CARs are winsorized at the 1st and 99th percentiles. The expected sign of β_1 is positive.

Table 5 provides the clearest summary of the intermediation gap. Equation (3) applied to the above/below median partition (Table 4) shows that firms above the median of log market capitalisation earned mean CARs of -0.08% at E1, while those below earned -2.57%, a statistically significant spread of $+2.49$ percentage points ($t = 3.82$, $N = 274$).

Table 5 reports the regression specification from Equation (4): in column (2), above-median enters at $+2.3$ pp ($p < 0.001$), friction at -0.31 pp per event ($p < 0.05$), and watch-list at $+5.2$ pp ($p < 0.10$), with an R^2 of 9.9%. At E3, the binary split reverses (Equation (3)): below-median firms gained $+4.10\%$ versus $+2.34\%$ for above-median (-1.75 pp, $p < 0.10$). In the E3 regression, the above-median coefficient in columns (6)–(9) is negative (-1.9 pp, $p < 0.05$ in column 7), while friction count has no explanatory power.

[17] See Law Commission of England and Wales, *Fiduciary Duties of Investment Intermediaries* (2014), https://lawcom.gov.uk/project/fiduciary-duties-of-investment-intermediaries/.

Table 4: Mean CARs by Size Group (E1 and E3)

	E1	E3
Below median	-2.57^{***}	$+4.10^{***}$
Above median	-0.08	$+2.34^{***}$
Difference	$+2.49^{***}$	-1.75^{*}
	(3.82)	(-1.79)
N	274	270

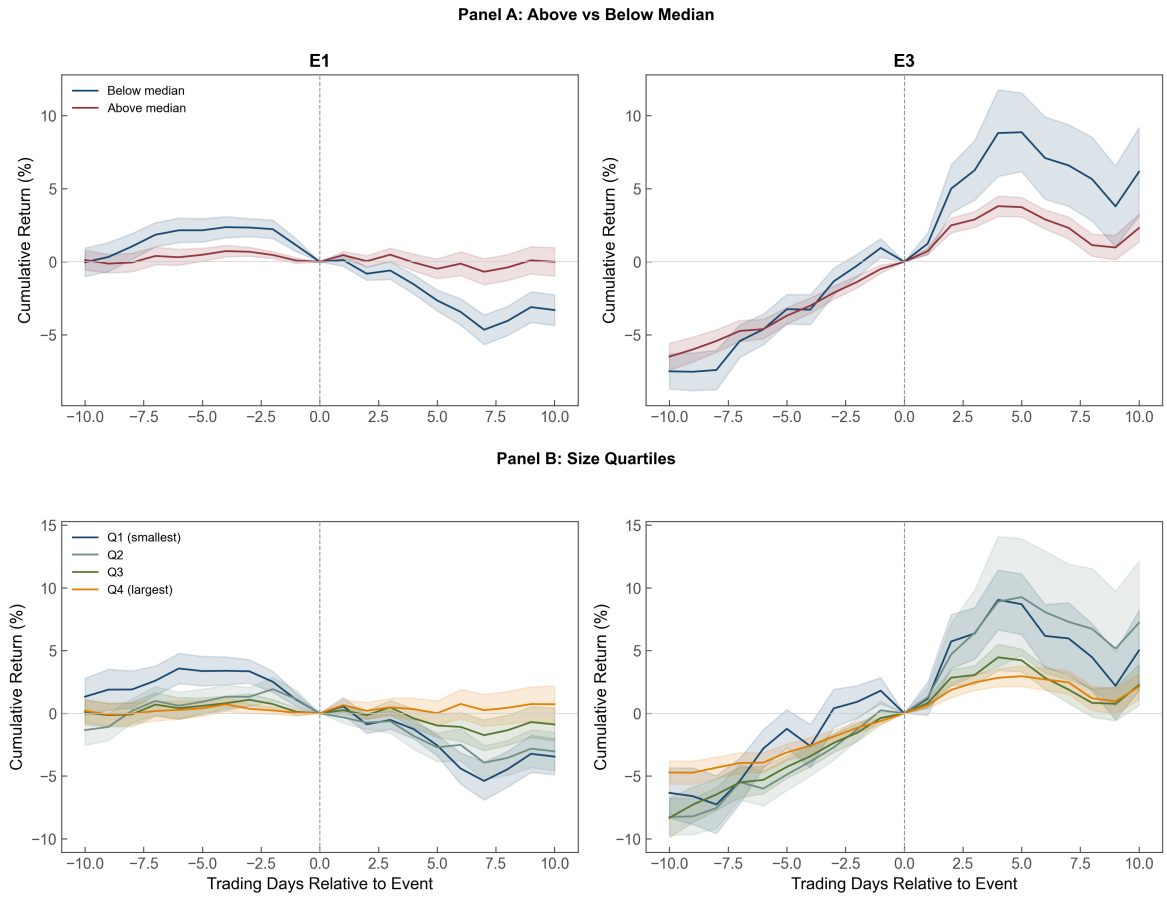
Notes: Mean cumulative abnormal returns (%) by above/below median log market capitalisation, estimated using a Fama–French three-factor model with Asia Pacific ex Japan factors over the $(-1, +3)$ trading-day window. t-statistic for the two-sample difference in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: Size and Announcement Returns: Regression Specifications (E1 and E3)

	E1					E3				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Above median	0.0249***	0.0227***	0.0211***	0.0175***	0.0203**	-0.0175*	-0.0190**	-0.0148	-0.0127	0.0025
	(0.0065)	(0.0062)	(0.0071)	(0.0067)	(0.0096)	(0.0098)	(0.0094)	(0.0102)	(0.0101)	(0.0117)
Friction count		-0.0031**	-0.0034**	-0.0031**	-0.0032**		0.0004	0.0013	0.0009	0.0015
		(0.0012)	(0.0016)	(0.0012)	(0.0016)		(0.0016)	(0.0019)	(0.0016)	(0.0015)
Watchlist		0.0524*	0.0575	0.0512	0.0559		-0.0612	-0.0763	-0.0126	0.0100
		(0.0316)	(0.0360)	(0.0327)	(0.0370)		(0.0697)	(0.0744)	(0.0692)	(0.0655)
Illiquidity				-0.0018	-0.0014				-0.0203*	-0.0253***
				(0.0098)	(0.0110)				(0.0106)	(0.0090)
Catalist				-0.0098	-0.0001				0.0199	0.0200
				(0.0126)	(0.0176)				(0.0192)	(0.0245)
Financial controls	No	No	Yes	No	No	No	No	Yes	No	No
Sector FE	No	No	No	No	Yes	No	No	No	No	Yes
N	274	274	240	272	243	270	270	238	268	248
R^2	0.051	0.099	0.135	0.099	0.205	0.013	0.023	0.051	0.043	0.246

Notes: OLS with HC1 robust standard errors in parentheses. The dependent variable is the firm-level cumulative abnormal return over the $(-1, +3)$ trading-day window around the event date, estimated using a Fama–French three-factor model with Asia Pacific ex Japan factors. Columns (1)–(5): E1; Columns (6)–(10): E3. Above median = 1 if $\ln(\text{MktCap})_i \geq$ sample median within each event. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 2: Cumulative abnormal return paths for above- and below-median firms across events E1 and E3, with 90% confidence bands.



Note: Panel A: Daily cumulative log returns for firms above (navy) vs. below (maroon) median log market capitalisation, normalised to zero at day 0. Panel B: Daily cumulative log returns by size quartile (Q1 = smallest). Left column: E1 (February 21, 2025). Right column: E3 (July 21, 2025). The E1-to-E3 reversal in the size gradient is visible in both panels.

The intermediation gap may narrow when the market receives credible information about mandate terms. At E3, three named fund managers were appointed with mandates requiring significant small- and mid-cap allocation. If this information is credible, the size effect should be smaller at E3 than at E1, or reversed in sign. The hypothesis yields two predictions:

$$\text{CAR}_{i,\text{E1}} = \alpha_1 + \beta_1 \text{AboveMedian}_i + \varepsilon_i \qquad \text{with } \beta_1 > 0 \tag{5}$$

$$\text{CAR}_{i,\text{E3}} = \alpha_2 + \beta_2 \text{AboveMedian}_i + \varepsilon_i \qquad \text{with } \beta_2 < \beta_1 \tag{6}$$

To test whether the change from β_1 to β_2 is statistically significant, we pool both events into a single stacked specification:

$$\text{CAR}_{i,e} = \alpha + \gamma_1 \text{AboveMedian}_i + \gamma_2 \text{AboveMedian}_i \times {}_{e=\text{E3}} + \delta_e + \varepsilon_{i,e} \tag{7}$$

where δ_e are event fixed effects and standard errors are clustered by stock code. Here γ_1 captures the size effect at E1 (corresponding to β_1), and γ_2 captures the *change* in the size effect from E1 to E3 (i.e., $\gamma_2 = \beta_2 - \beta_1$). A negative γ_2 indicates that the size advantage narrows at implementation.

The E1-to-E3 reversal is the key test of the intermediation gap hypothesis (Equations (5) and (6)). The reversal documented in Table 5 is visible in the CAR trajectories of Figure 2: at E1, below-median firms drift downward while above-median firms are flat; at E3 the pattern inverts. This sign flip is the core evidence for the intermediation gap hypothesis; the stacked specification in Equation (7) confirms that $\gamma_2 < 0$. A paired-date reversal placebo confirms that the flip is statistically unusual. For the primary above-median specification, only 2% of the 200 random date-pairs produce a reversal as large as the observed $\Delta\hat{\beta} = -0.042$ (permutation $p = 0.030$); for the continuous $\ln(\text{MktCap})_i$ specification, only 2% do ($\Delta\hat{\beta} = -0.011$, $p = 0.030$; Section A.1.1). Table 20 in the Appendix confirms the reversal parametrically across five alternative size measures, all of which exhibit the E1-positive/E3-negative pattern.

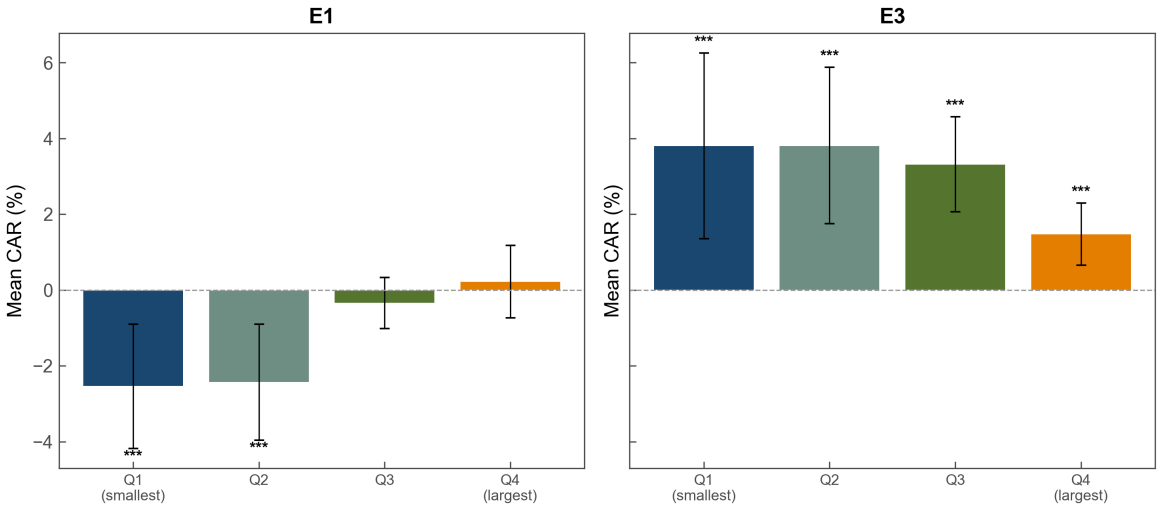
The size effect is also examined non-parametrically by sorting firms into quartiles of market capitalisation within each event (Equation (3)). The quartile analysis in Table 6 confirms that the effect operates monotonically across the full size distribution: at E1, the smallest quartile experienced mean CARs of −2.54% ($p < 0.01$) while the largest quartile gained +0.22% (insignificant), producing a $Q4 - Q1$ spread of +2.76 percentage points ($p < 0.01$). At E3, the gradient reverses: $Q1 = +3.83\%$ ($p < 0.01$), $Q4 = +1.48\%$ ($p < 0.01$), with the spread turning negative (−2.36, $p < 0.10$). Using continuous $\ln(\text{MktCap})_i$ as a robustness check yields a coefficient of +0.0054 per log-unit ($p < 0.01$; Table 21).

Table 6: Size Quartile CARs Across Events

Quartile	E1	E2	E3
Q1 (smallest)	-2.54^{***}	−0.04	$+3.83^{***}$
Q2	-2.43^{***}	−0.01	$+3.83^{***}$
Q3	−0.34	+0.18	$+3.32^{***}$
Q4 (largest)	+0.22	−0.50	$+1.48^{***}$
$Q4 - Q1$	$+2.76^{***}$	−0.46	-2.36^{*}
N	272	273	269

Notes: Mean cumulative abnormal returns (%) estimated over the $(-1, +3)$ trading-day window around the event date using a Fama–French three-factor model with Asia Pacific ex Japan factors. Size quartiles formed on log market capitalisation within each event. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 3: Mean cumulative abnormal returns by market capitalisation quartile at events E1 and E3. Error bars denote 90% confidence intervals.



Note: Mean FF3 CAR$(-1, +3)$ by market capitalisation quartile. Error bars show 90% confidence intervals. At E1, returns increase monotonically with size ($Q4 - Q1 = +2.76$ pp, $p < 0.01$). At E3, the gradient reverses: smaller firms gain more ($Q1 = +3.83\%$, $Q4 = +1.48\%$), consistent with the mandate design targeting small- and mid-cap allocation.

Figure 3 complements the quartile estimates in Table 6, showing the monotonic gradient at E1 ($Q1 = -2.54\%$, $Q4 = +0.22\%$) and its reversal at E3.

Table 7 examines two additional dimensions of fund manager investability. Panel A sorts firms by I/B/E/S analyst coverage. At E1, firms with zero analyst coverage (those invisible to the institutional investment ecosystem) experienced mean CARs of -1.80% ($p < 0.01$), while firms covered by six or more analysts posted -0.42% (insignificant). At E3, when the GEMS research expansion specifically targeted under-covered firms, the gradient reversed: zero-coverage firms gained $+3.04\%$ while highly covered firms gained only $+1.05\%$ (Spearman $\rho = -0.182$, $p = 0.060$). Panel B reports foreign institutional ownership quartile CARs at E1: firms in the top quartile gained $+0.20\%$, while those in the bottom quartile lost -2.36% (Spearman $\rho = +0.129$, $p = 0.083$). The gradients in both coverage and foreign ownership parallel the size gradient, with the ownership correlation reaching marginal significance. In multivariate specifications, neither analyst coverage nor foreign institutional ownership retains significance after controlling for log market capitalisation, indicating that these variables capture the same underlying investability construct rather than independent channels. This strong collinearity precludes their joint inclusion as regressors; accordingly, the horse race specification in Table 13 uses the above-median size indicator as the sole investability proxy. The pattern is consistent with the information asymmetry mechanism identified by Didier (2011), who shows that foreign institutional investors concentrate in large firms precisely because size proxies for the information environment that enables institutional participation.

Table 7: Analyst Coverage and Foreign Institutional Ownership

Panel A: Mean CARs (%) by I/B/E/S Analyst Coverage Group

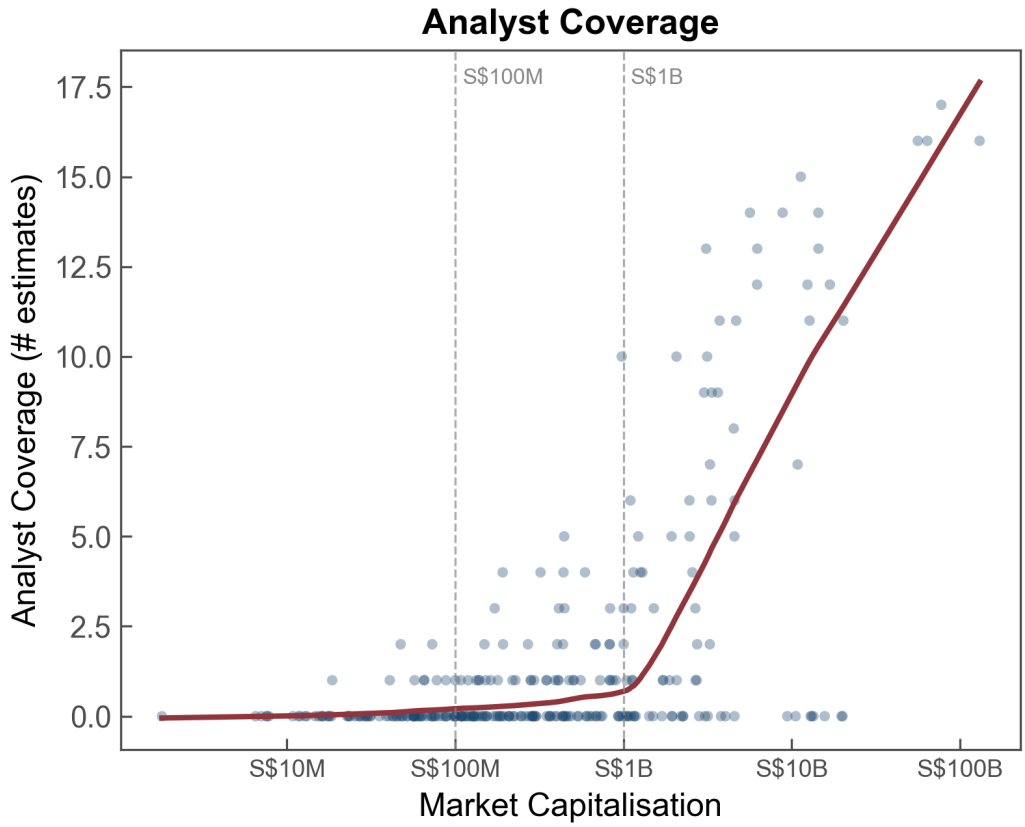
	E1 (Feb 2025)		E3 (Jul 2025)	
Coverage group	Mean CAR	N	Mean CAR	N
Zero coverage	-1.80^{***}	165	$+3.04^{***}$	163
1–2 analysts	-0.65	57	$+2.81^{***}$	51
3–5 analysts	-0.23	21	$+3.50^{**}$	22
6+ analysts	-0.42	31	$+1.05^{**}$	34
Spearman ρ (covered only)	-0.083	109	-0.182^{*} ($p = 0.060$)	107

Panel B: Foreign Institutional Ownership Quartile CARs (E1 only)

	Mean CAR	Mean $\mathrm{IO}_{\mathrm{for}}$	N
Q1 (lowest)	-2.36^{**}	0.01%	46
Q2	-0.67	0.23%	44
Q3	-0.85	1.35%	45
Q4 (highest)	$+0.20$	9.72%	46
Spearman ρ	$+0.129^{*}$ ($p = 0.083$)		181

Notes: Panel A: Mean CARs (%) by I/B/E/S analyst coverage group. Firms not in I/B/E/S are classified as zero coverage. Spearman ρ computed on covered firms only. Panel B: E1 only. Foreign institutional ownership ($\mathrm{IO}_{\mathrm{for}}$) from FactSet, most recent quarter $\leq$ event date (Q4 2024). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 4: Analyst coverage by market capitalisation. Dashed lines mark approximate thresholds at S$100 million and S$1 billion.



Note: Each point represents one SGX-listed firm. The I/B/E/S analyst estimate count rises sharply above approximately S$1 billion in market capitalisation. Dashed lines at S$100 million and S$1 billion mark approximate thresholds.

Figure 4 visualises the analyst-coverage cliff documented in Table 7: below roughly S$1 billion in market capitalisation, sell-side coverage is near zero, reinforcing the interpretation that the size effect in Table 20 reflects an investability constraint rather than a smooth risk gradient.

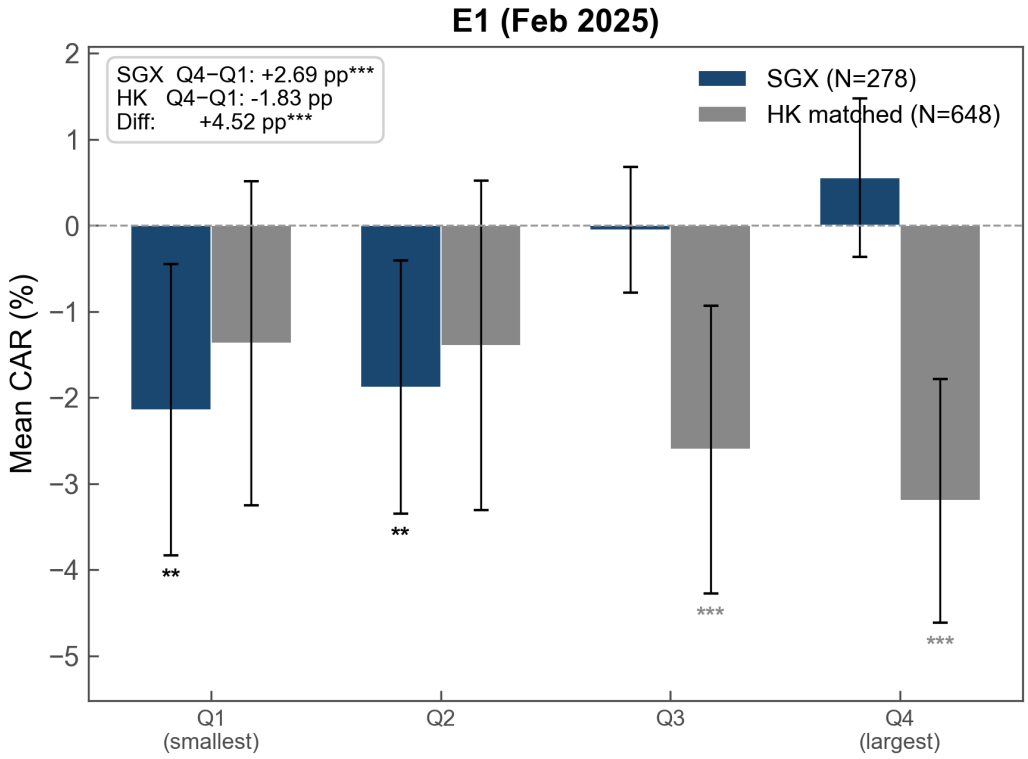
The SGX equity universe, with a total market capitalisation of just under S$1 trillion, exhibits extreme size dispersion, ranging from micro-cap firms valued at under S$2 million to blue-chip constituents exceeding S$130 billion. In a market with this degree of dispersion, firm size does not merely correlate with other characteristics—it largely determines them. Firms below roughly S$100 million in capitalisation typically have no sell-side analyst coverage, negligible foreign institutional ownership, and minimal daily trading volume. Firms above S$1 billion, by contrast, have all three. The gradient between these two states is steep, and for practical purposes resembles a threshold rather than a continuum.

In a deep market like the United States, government capital channelled through fund managers can reach mid-sized companies because those companies already sit above the visibility threshold; the fund manager's task is to overweight them slightly. In Singapore, the intended beneficiaries sit below the threshold entirely—absent from the investable universe, not merely underweighted

within it. The intermediation gap follows directly from the institutional architecture of professional asset management: compliance screens, liquidity requirements, benchmark constraints, and due diligence obligations that collectively render small SGX-listed firms inaccessible regardless of the policy mandate.

A natural concern is that the E1 size gradient reflects a concurrent regional size rotation rather than a Singapore-specific treatment effect. To address this, we construct a matched Hong Kong placebo: each SGX firm is matched to its three nearest Hong Kong Exchange (HKEX) neighbours by log market capitalisation, producing a sample of 644 Hong Kong firms, and we estimate the identical size–CAR specification on both markets at E1. Figure 5 visualises the contrast. SGX exhibits the monotonic upward gradient documented above; the matched HK sample shows a significant negative size gradient ($Q4 - Q1 = -2.78$ percentage points, $p = 0.024$), opposite in sign to the SGX pattern. The cross-market difference in the binary above-median coefficient is significant ($p < 0.001$), ruling out a regional size rotation as an alternative explanation. Full regression results appear in Table 17.

Figure 5: Cross-Market Placebo: Size Quartile CARs at E1 (SGX vs Matched HK)



Note: Bars show mean FF3 CARs (%) by size quartile at E1 (February 2025). Navy bars: SGX firms. Gray bars: HKEX firms matched by log market capitalisation ($K = 3$). Full regression results in Table 17. Sample sizes and coefficient values in the figure reflect an earlier matching procedure; the regression table reports the authoritative estimates.

The intermediation gap hypothesis addresses the reform's first instrument: capital deployment through professional fund managers. The E1 announcement, however, bundled capital deployment

with a second instrument: deregulation of SGX's post-listing oversight regime. If the market prices these instruments through distinct institutional channels, the reform should generate cross-sectional variation not only along the size dimension (reflecting intermediation constraints) but also along a friction dimension (reflecting the expected information consequences of reduced oversight). The following hypothesis examines this second channel.

The size results establish that the reform's capital instrument was priced through the lens of fund manager constraints. We now turn to the reform's second instrument, deregulation, and ask whether the proposed removal of SGX's oversight apparatus was priced through a structurally distinct channel.

5.3 Friction, Distress, and the Watch-List

Regulatory friction (accumulated queries, trading halts, and suspensions) is conventionally understood as a cost imposed on firms by the regulator. Under this view, reforms that reduce friction should benefit high-friction firms the most. An alternative interpretation holds that the market values the information content of regulatory interventions: queries signal that the regulator has identified potential problems, and halts and suspensions signal that those problems are serious.

If the market views regulatory friction as informative monitoring rather than arbitrary burden, removing it through deregulation should be priced *negatively* for high-friction firms, because investors lose a source of discipline. The logic has an adverse selection dimension: when the regulator publicly queries a firm, the market learns which firms have potential problems; when the query regime is removed, investors can no longer distinguish clean firms from problematic ones that are no longer flagged, creating a lemons problem that depresses valuations across the high-friction segment.

The specification is:

$$\text{CAR}_{i,\text{E1}} = \alpha + \beta_1 \text{FricCount}_i + \beta_2 \text{Watchlist}_i + \beta_3 \text{AboveMedian}_i + \gamma' X_i + \varepsilon_i \tag{8}$$

where FricCount_i is the total count of regulatory friction events (queries, halts, suspensions) for firm i over the pre-event period, Watchlist_i indicates inclusion on SGX's financial watch-list, and X_i is a vector of financial controls. The expected sign of β_1 is negative (friction penalised) and of β_2 positive (stigma relief), reflecting the distinction between general friction (informative monitoring whose removal is costly) and specific removable stigma (an identifiable label whose removal has discrete value).

Estimating Equation (8), friction count enters negatively and significantly in the univariate specification (Table 8, Spec 1): each additional regulatory friction event is associated with -0.34 percentage points of abnormal return ($p < 0.01$). Adding the watch-list indicator and above-median size control in Spec 2, friction retains its sign (-0.0031, $p < 0.05$), the watch-list coefficient is positive and marginally significant ($+0.052$, $p < 0.10$), and the above-median indicator enters at $+0.023$ ($p < 0.001$). Adding financial controls (ROA, leverage, book-to-market) in Spec 3, friction

is -0.0034 ($p < 0.05$), with the sign and economic magnitude unchanged. Adding illiquidity and Catalist listing status (Spec 4) yields friction at -0.0031 ($p < 0.05$); including SSIC sector fixed effects (Spec 5) produces a similar estimate (-0.0032, $p < 0.05$). The friction penalty is consistently negative and statistically significant across all five specifications.

Table 8: Friction and Announcement Returns (E1)

	FF3					CAPM
	(1)	(2)	(3)	(4)	(5)	(6)
Friction count	-0.0034***	-0.0031**	-0.0034**	-0.0031**	-0.0032**	-0.0036***
	(0.0013)	(0.0012)	(0.0016)	(0.0012)	(0.0016)	(0.0011)
Watchlist		0.0524*	0.0575	0.0512	0.0559	0.0323**
		(0.0316)	(0.0360)	(0.0327)	(0.0370)	(0.0127)
Above median		0.0227***	0.0211***	0.0175***	0.0203**	0.0263***
		(0.0062)	(0.0071)	(0.0067)	(0.0096)	(0.0058)
Illiquidity				-0.0018	-0.0014	
				(0.0098)	(0.0110)	
Catalist				-0.0098	-0.0001	
				(0.0126)	(0.0176)	
Financial controls	No	No	Yes	No	No	No
Sector FE	No	No	No	No	Yes	No
N	274	274	240	272	243	274
R^2	0.049	0.099	0.135	0.099	0.205	0.140

Notes: OLS with HC1 robust standard errors in parentheses. The dependent variable is the firm-level cumulative abnormal return over the $(-1, +3)$ trading-day window around E1 (February 21, 2025). Columns (1)–(5): Fama–French three-factor model (Asia Pacific ex Japan). Column (6): market model (CAPM) with SGX equal-weighted index, presented jointly because the S$40 million watch-list threshold is a market-cap criterion and the FF3 SMB factor mechanically absorbs watch-list variation. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The sign contrast within Table 8 is striking: friction count is negative (information loss) while watch-list status is positive (stigma relief). Under FF3 (column 2), the watch-list coefficient is $+0.052$ ($p < 0.10$); under CAPM (column 6), it strengthens to $+0.032$ ($p < 0.05$). The attenuation under FF3 reflects the SMB factor absorbing variation linked to the S$40 million market capitalisation threshold that defines watch-list eligibility. Under CAPM, which does not include a size factor, the full institutional channel is visible: each additional friction event predicts -0.36 percentage points ($p < 0.01$), while watch-list status predicts $+3.2$ percentage points ($p < 0.05$).

We also tested whether three consecutive pre-tax losses, the watch-list trigger criterion, independently predict E1 CARs; the coefficient is negative and marginally significant (-0.043, $p < 0.10$),

consistent with fundamental impairment being penalised, but not robust to additional controls.[18] Trading suspension history enters negatively but is not significant. The positive watch-list coefficient, combined with its directional consistency in the E4 implementation event (Table 9), supports the interpretation that the market distinguishes between the watch-list (a specific and removable stigma) and other forms of regulatory friction.

The watch-list was formally abolished on October 29, 2025, with all issuers automatically removed. This provides a test of the stigma-relief mechanism at implementation rather than announcement:

$$\mathrm{CAR}_{i,\mathrm{E4}} = \alpha + \beta_1 \mathrm{Watchlist}_i + \beta_2 \ln(\mathrm{MktCap})_i + \gamma' X_i + \varepsilon_i \tag{9}$$

where $\mathrm{Watchlist}_i$ reflects pre-removal watch-list status. A positive β_1 would confirm that the market prices watch-list removal as stigma relief at implementation.

The E4 evidence supports the stigma-relief mechanism, though precision is limited by the small treatment group. When the watch-list was formally abolished on October 29, 2025, applying Equation (3), the five previously watch-listed firms gained +12.7% on average under FF3, compared to +1.0% for non-watch-listed firms (Table 9, Panel A). The difference of +11.8 percentage points is economically large, but the Welch t-test p-value of 0.279 reflects the small number of treated firms ($N = 5$). Under CAPM, the pattern is similar: +7.3% for watch-listed versus −1.1% for non-watch-listed ($p = 0.363$). Panel B reports the regression from Equation (9): under FF3, the watch-list coefficient is +0.118 in the bare specification and +0.111 with the above-median control (columns 1–2); under CAPM, +0.084 and +0.082 (columns 3–4). No other cross-sectional variable predicts E4 CARs, consistent with the event being narrowly targeted at watch-list removal.

[18] The friction × distress interaction coefficient is +0.005 ($p = 0.12$), robust across factor models (CAPM, FF3, FF5) and sample specifications.

Table 9: Watch-List Removal Implementation (E4)

Panel A: Mean CARs by watchlist status				
	FF3		CAPM	
	Mean CAR	p-value	Mean CAR	p-value
Watchlisted (N=5)	0.1273	0.247	0.0731	0.423
Non-watchlisted (N=253)	0.0098***	0.007	-0.0111***	0.002
Difference	0.1176	0.279	0.0841	0.363
Panel B: Regression				
	FF3		CAPM	
	(1)	(2)	(3)	(4)
Watchlist	0.1176	0.1110	0.0841	0.0817
	(0.0844)	(0.0847)	(0.0737)	(0.0740)
Above median		-0.0129*		-0.0048
		(0.0072)		(0.0071)
N	258	258	258	258
R^2	0.063	0.073	0.036	0.038

Notes: CARs estimated over the $(-1, +3)$ trading-day window around E4 (October 29, 2025). Panel A: Mean CARs and p-values for H_0: mean $= 0$ (individual groups) or H_0: diff $= 0$ (Welch t-test), under both FF3 (Fama–French three-factor, Asia Pacific ex Japan) and CAPM (market model, SGX equal-weighted index). Panel B: OLS with HC1 robust standard errors in parentheses. Columns (1)–(2): FF3. Columns (3)–(4): CAPM. CAPM presented jointly because the S$40 million watch-list threshold is a market-cap criterion and the FF3 SMB factor mechanically absorbs watch-list variation. Panel A includes all firms with valid CARs ($N = 263$); Panel B regression sample is $N = 258$ after excluding firms with missing control variables. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.3.1 Friction Type Decomposition

The aggregate friction count treats all regulatory interventions equally. But queries (information-seeking requests from the regulator), halts and suspensions (interruptions to trading), and regulatory actions (enforcement proceedings) carry different informational content. If the market values regulatory friction as a monitoring signal, the effect should concentrate in the friction types that convey the most serious information about firm-level compliance problems.

We decompose the friction count into three components entered simultaneously:

$$\text{CAR}_{i,\text{E1}} = \alpha + \beta_1 \text{QueryFric}_i + \beta_2 \text{HaltFric}_i + \beta_3 \text{RegFric}_i + \gamma' X_i + \varepsilon_i \tag{10}$$

where QueryFric_i counts query-type friction events, HaltFric_i counts halt and suspension-type events, and RegFric_i counts regulatory action-type events.

Table 10 reports the results. Halt and suspension-type friction drives the entire aggregate effect: the coefficient is -0.0036 ($p < 0.01$) in the univariate specification, -0.0038 ($p < 0.01$) with channel controls, and -0.0036 ($p < 0.05$) with financial controls. Query-type friction is statistically indistinguishable from zero across all specifications (-0.0025 to -0.0013, all $p > 0.10$). Regulatory-action-type friction enters with a large negative coefficient (-0.013 to -0.043) but is too rare for reliable inference (97% of firms have zero regulatory actions). The market thus prices the removal of oversight that interrupts trading to signal serious compliance problems; routine regulatory questioning is not priced as informative monitoring. This sharpens the information channel interpretation: it is the severity of the regulatory signal, not its frequency, that the market values.

Table 10: Friction Type Decomposition (E1)

	(1)	(2)	(3)
Query-type friction	-0.0025	-0.0011	-0.0013
	(0.0023)	(0.0022)	(0.0025)
Halt/suspension-type friction	-0.0036***	-0.0038***	-0.0036**
	(0.0013)	(0.0012)	(0.0018)
Regulatory-type friction	-0.0127	-0.0236	-0.0433
	(0.0258)	(0.0275)	(0.0381)
Above median		0.0241***	0.0227***
		(0.0064)	(0.0073)
Watchlist		0.0627*	0.0818*
		(0.0360)	(0.0460)
Financial controls	No	No	Yes
N	274	274	240
R^2	0.056	0.112	0.148

Notes: OLS with HC1 robust standard errors in parentheses. The dependent variable is the firm-level cumulative abnormal return over the $(-1, +3)$ trading-day window around E1, estimated using a Fama–French three-factor model with Asia Pacific ex Japan factors. Query-type: SGX public queries. Halt/suspension-type: trading halts and suspensions. Regulatory-type: formal regulatory actions. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.3.2 Friction Recency

Does the market price recent regulatory scrutiny differently from historical? If friction serves as a flow signal about current compliance risk, recent events should carry more weight. Alternatively, if friction proxies for a stock of accumulated reputation, older events should drive the effect.

We decompose friction into recent (within 12 months of E1, i.e., March 2024 to January 2025) and

old (pre-March 2024):

$$\text{CAR}_{i,\text{E1}} = \alpha + \beta_1 \text{RecentFric}_i + \beta_2 \text{OldFric}_i + \gamma' X_i + \varepsilon_i \tag{11}$$

Table 11 shows that old friction drives the aggregate effect: the coefficient is -0.0046 ($p < 0.10$) in the univariate specification and -0.0044 ($p < 0.10$) with channel controls. Recent friction is statistically indistinguishable from zero (-0.0032 and -0.0026, both $p > 0.10$), even slightly less negative than the old-friction coefficient. The market thus treats regulatory friction as accumulated reputation rather than current scrutiny. The information channel reflects a stock of regulatory attention the market has relied on over time, not a flow of recent events. This is consistent with the adverse selection interpretation: firms whose regulatory history the market has learned to use as a screening device are the ones penalised when that screening device is proposed for removal.

Table 11: Friction Recency (E1)

	(1)	(2)
Friction count (recent)	-0.0032	-0.0026
	(0.0032)	(0.0030)
Friction count (old)	-0.0046*	-0.0044*
	(0.0027)	(0.0025)
Above median		0.0221***
		(0.0062)
Watchlist		0.0541*
		(0.0325)
N	274	274
R^2	0.060	0.108

Notes: OLS with HC1 robust standard errors in parentheses. The dependent variable is the firm-level cumulative abnormal return over the $(-1, +3)$ trading-day window around E1, estimated using a Fama–French three-factor model with Asia Pacific ex Japan factors. Recent friction: events from March 2024 to January 2025. Old friction: events before March 2024. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.3.3 Stacked E1–E3 Interaction

The reversal placebo (Section A.1.1) tests whether the E1-to-E3 sign flip in the size coefficient is statistically unusual; a complementary question is whether the two channels exhibit *different* cross-event dynamics. Pooling E1 and E3 into a stacked specification with event fixed effects:

$$\text{CAR}_{i,e} = \alpha + \beta_1 \text{FricCount}_i + \beta_2 \text{AboveMedian}_i + \gamma_1 \text{FricCount}_i \times_{e=\text{E3}} + \gamma_2 \text{AboveMedian}_i \times_{e=\text{E3}} + \delta_e + \varepsilon_{i,e} \tag{12}$$

with standard errors clustered by stock code. γ_1 tests whether the friction penalty changes from E1 to E3; γ_2 tests whether the size advantage changes.

Table 12 confirms that the two channels have structurally different cross-event dynamics. The size channel reverses: $\gamma_2 = -0.039$ ($p < 0.001$) in the specification with channel controls, indicating that the above-median advantage at E1 flips to a disadvantage at E3 once mandate terms are disclosed. The friction channel, by contrast, does not reverse: $\gamma_1 = +0.003$ (insignificant), indicating that the friction penalty simply disappears at E3 rather than changing sign. This contrast is informative. Size reverses because the E3 mandate terms changed the market's expectation about where capital would flow. Friction goes to zero because there was no new friction-relevant information at E3; the deregulatory content of E3 was negligible compared to the capital-deployment signal. Two channels, two different cross-event dynamics, consistent with genuinely independent pricing dimensions.

Table 12: Stacked E1–E3 Interaction

	(1)	(2)	(3)
Friction count	-0.0034***	-0.0029**	-0.0033**
	(0.0013)	(0.0012)	(0.0016)
Above median		0.0214***	0.0232***
		(0.0061)	(0.0071)
Post-E3	0.0345***	0.0572***	0.0557***
	(0.0065)	(0.0100)	(0.0112)
Friction × Post-E3	0.0040	0.0031	0.0044
	(0.0026)	(0.0025)	(0.0032)
Above median × Post-E3		-0.0387***	-0.0397***
		(0.0107)	(0.0110)
Financial controls	No	No	Yes
N	544	544	478
R^2	0.118	0.137	0.155

Notes: OLS with event fixed effects and standard errors clustered by stock code. The dependent variable is the firm-level cumulative abnormal return over the $(-1, +3)$ trading-day window, estimated using a Fama–French three-factor model with Asia Pacific ex Japan factors. Post-E3 = 1 for E3 observations. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.4 Channel Separation

The preceding subsections posit that the bundled reform operates through two structurally distinct channels: capital deployment, whose market impact is mediated by fund manager investability constraints, and deregulation, whose impact is mediated by the information content of the oversight

regime. If these are genuinely independent dimensions rather than alternative measures of a single underlying characteristic (small firms tend to be both uninvestable and high-friction), they should load simultaneously without either absorbing the other:

$$\text{CAR}_{i,\text{E1}} = \alpha + \beta_1 \text{AboveMedian}_i + \beta_2 \text{FricCount}_i + \beta_3 \text{Watchlist}_i + \gamma' X_i + \lambda_s + \varepsilon_i \tag{13}$$

where AboveMedian_i equals one if a firm's log market capitalisation exceeds the within-event-sample median, X_i is a vector of controls (illiquidity, Catalist board tier, and, in augmented specifications, financial controls and board composition variables), and λ_s denotes SSIC level-1 sector fixed effects. The hypothesis is that $\beta_1 > 0$ and $\beta_2 < 0$ simultaneously, with both surviving the inclusion of the other. Table 13 reports specifications with progressively saturated controls.

Table 13 reports the horse race specification from Equation (13). When entered jointly (Spec 3), the above-median indicator loads at +0.018 ($p < 0.01$) and friction count at −0.0031 ($p < 0.05$), with neither absorbing the other. The watch-list indicator is positive (+0.051) but imprecisely estimated, while Amihud illiquidity and Catalist board tier are statistically indistinguishable from zero. The joint specification achieves an R^2 of 9.9%, with explanatory power concentrated in the intermediation and information channels. Adding SSIC level-1 sector fixed effects (Spec 4) raises R^2 to 20.5% while preserving both the above-median indicator (+0.020, $p < 0.05$) and friction (−0.0032, $p < 0.05$), ruling out sectoral confounds. Augmenting with governance variables (Spec 5) confirms that board independence share (−0.020, insignificant) and CEO-chair separation (+0.009, insignificant) do not constitute a third independent dimension, while the above-median (+0.019, $p < 0.01$) and friction (−0.003, $p < 0.05$) coefficients remain unchanged. Governance quality, as measured by board structure, does not constitute a third independent dimension of announcement-day pricing. Using continuous $\ln(\text{MktCap})_i$ in place of the binary indicator yields qualitatively identical results but with a wider confidence interval ($p = 0.13$; Table 21). The simultaneous survival of both coefficients confirms that the reform was priced on two genuinely independent dimensions: intermediation (mediated by investability constraints) and information (mediated by the oversight regime).

Table 13: Channel Separation (E1)

	(1)	(2)	(3)	(4)	(5)
Above median	0.0249***		0.0175***	0.0203**	0.0193***
	(0.0065)		(0.0067)	(0.0096)	(0.0072)
Friction count		-0.0034***	-0.0031**	-0.0032**	-0.0030**
		(0.0013)	(0.0012)	(0.0016)	(0.0013)
Watchlist			0.0512	0.0559	0.0508
			(0.0327)	(0.0370)	(0.0329)
Illiquidity (z-Amihud)			-0.0018	-0.0014	-0.0012
			(0.0098)	(0.0110)	(0.0099)
Catalist			-0.0098	-0.0001	-0.0055
			(0.0126)	(0.0176)	(0.0129)
Independence share					-0.0195
					(0.0244)
No CEO duality					0.0085
					(0.0102)
Sector FE	No	No	No	Yes	No
N	274	274	272	243	260
R^2	0.051	0.049	0.099	0.205	0.096

Notes: OLS with HC1 robust standard errors in parentheses. The dependent variable is the firm-level cumulative abnormal return over the $(-1, +3)$ trading-day window around E1 (February 21, 2025), estimated using a Fama–French three-factor model with Asia Pacific ex Japan factors. Above median = 1 if $\ln(\text{MktCap})_i \geq$ sample median. Spec (4) includes SSIC level-1 sector fixed effects. Spec (5) adds board independence share and CEO-chair separation. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.5 Conditional Liquidity

Although illiquidity enters the channel separation specification as a control, the theory implies that its effect should be conditional rather than global. Within the subset of large-cap firms (those that clear the fund manager investability screen), illiquidity may determine which specific firms are selected for purchase. This conditional effect reflects a two-stage selection process: fund managers first screen on size, then on tradability within the investable set. The specification is estimated within each size tercile k:

$$\text{CAR}_{i,\text{E1}} = \alpha_k + \beta_{1,k}\,\text{Illiq}_i + \varepsilon_i \qquad \text{for } i \in \text{Tercile}_k,\ k \in \{\text{Small, Medium, Large}\} \tag{14}$$

with the expectation that $\beta_{1,\text{Large}} < 0$ and $\beta_{1,\text{Small}} \approx 0$.

Illiquidity does not appear as an independent predictor of abnormal returns in unconditional specifications at any event. However, Equation (14) reveals a notable conditional pattern when estimated within size terciles. At E1 (Table 14, Panel A), the Amihud illiquidity coefficient within the large-cap tercile is $\hat{\beta}_{1,\text{Large}} = -1.78$ ($p < 0.01$): among the firms that clear the fund manager investability screen, more illiquid stocks experienced significantly more negative CARs. The medium and small tercile coefficients are economically negligible and statistically indistinguishable from zero (-0.044 and -0.001, both $p > 0.10$). This pattern is consistent with a two-stage selection process in which fund managers first screen on size and then, within the investable set, screen on tradability.

At E5 (Panel B), the conditional pattern reverses. The small-cap tercile coefficient is positive and significant ($+0.009$, $p < 0.05$): among the smallest firms, less liquid stocks experienced more positive CARs. The medium and large tercile coefficients are indistinguishable from zero.

The sign flip reflects a difference in policy mechanism. At E1, capital deployment through institutional managers penalised illiquidity within the investable set: fund managers screening on tradability would avoid the most illiquid large-cap stocks. At E5, the policy instrument changed. Board lot reduction and designated market maker incentives directly address the liquidity deficit: reducing minimum transaction sizes lowers trading costs disproportionately for illiquid stocks, and guaranteed market-maker quotes move thinly traded securities from intermittently tradeable to continuously tradeable. The same variable, opposite sign, because E1 channelled capital through existing market structure (which penalises illiquidity) while E5 proposed to change the market structure itself (which rewards the most illiquid).

The master coefficients table (Table 15) reveals a tension at E5: the above-median coefficient is positive and significant ($+0.015$, $p < 0.01$), meaning the overall event still favoured larger firms, even as illiquidity carried a positive coefficient ($+0.007$, $p < 0.05$). This likely reflects the bundling of distinct mechanisms within E5: the dual listing bridge (which carries fixed costs limiting its use to larger firms) and the additional S$2.85 billion in EQDP capital reinforced the intermediation channel, while the market structure enhancements simultaneously benefited illiquid firms. E5 thus combined two policy instruments with opposing liquidity implications, producing simultaneous positive coefficients on both size and illiquidity. Table 14 reports the tercile-stratified results for both events.

Table 14: Conditional Liquidity Effects by Size Tercile (E1 and E5)

	Small	Medium	Large
Panel A: E1 (February 2025)			
Illiquidity (std.)	-0.0008	-0.0435	-1.7785***
	(0.0111)	(0.1355)	(0.6095)
N	91	89	91
Panel B: E5 (November 2025)			
Illiquidity (std.)	0.0089**	0.0005	-0.0021
	(0.0041)	(0.0044)	(0.0259)
N	86	85	86

Notes: Panel A: E1 (February 21, 2025). Panel B: E5 (November 19, 2025). OLS with HC1 robust standard errors in parentheses. The dependent variable is the firm-level cumulative abnormal return over the $(-1, +3)$ trading-day window, estimated using a Fama–French three-factor model with Asia Pacific ex Japan factors. Size terciles formed on log market capitalisation within event. Illiquidity (std.) is the standardised Amihud illiquidity measure from the pre-event cross-section. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 15 consolidates the horse race specification across all five events using the above-median size indicator. The intermediation and information channels are specific to E1: the above-median coefficient is significant at E1 (+0.018, $p < 0.01$) and at E5 (+0.015, $p < 0.01$), while friction count is significant only at E1 (-0.003, $p < 0.05$). No cross-sectional variable consistently predicts CARs at E2 or E3. The watch-list coefficient is large and positive at E4 (+0.094) but imprecisely estimated due to the small treatment group. Illiquidity is marginally significant at E3 (-0.020, $p < 0.10$); the E5 illiquidity sign flip is discussed in Section 5.5.

Table 15: Master Coefficient Table: Horse Race Across Events

	E1	E2	E3	E4	E5
Above median	0.0175***	0.0003	-0.0127	-0.0060	0.0150***
	(0.0067)	(0.0069)	(0.0101)	(0.0093)	(0.0053)
Friction count	-0.0031**	0.0004	0.0009	-0.0006	0.0007
	(0.0012)	(0.0011)	(0.0016)	(0.0010)	(0.0009)
Watchlist	0.0512	0.0456	-0.0126	0.0941	0.0000
	(0.0327)	(0.0616)	(0.0692)	(0.0780)	(0.0163)
Illiquidity	-0.0018	-0.0072	-0.0203*	0.0065	0.0074**
	(0.0098)	(0.0092)	(0.0106)	(0.0073)	(0.0033)
Catalist	-0.0098	0.0072	0.0199	-0.0091	-0.0048
	(0.0126)	(0.0143)	(0.0192)	(0.0144)	(0.0083)
N	272	273	268	258	258
R^2	0.099	0.008	0.043	0.091	0.093
Mean CAR	-0.0132	-0.0011	0.0312	0.0120	0.0062

Notes: OLS with HC1 robust standard errors in parentheses. The dependent variable is the firm-level cumulative abnormal return over the $(-1, +3)$ trading-day window, estimated using a Fama–French three-factor model with Asia Pacific ex Japan factors. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6 Limitations

The paper's identification strategy relies on cross-sectional variation within single events rather than on coefficient differences across events with distinct policy content. The research design originally intended to exploit the difference between E2 (deregulation-specific) and E3 (capital-deployment-specific) to identify each channel through cross-event variation. This separation was only partially achieved: E2 produced null mean abnormal returns ($p = 0.721$), consistent with the consultation having been fully anticipated at E1, while E3 produced uniformly positive mean returns but still exhibited a reversed size gradient (size quartile Q1 = +3.83%, Q4 = +1.48%) rather than a clean level shift attributable to capital deployment alone. The paper therefore relies

primarily on within-E1 decomposition, showing that size and friction load simultaneously, rather than the cleaner identification that full cross-event variation would have provided.

Size, analyst coverage, and foreign institutional ownership are highly correlated dimensions of a single underlying construct that we term *investability*: the degree to which a firm is visible to, and tradeable by, professional fund managers. Although Table 7 shows that low-coverage and low-ownership firms earned lower abnormal returns at E1, group means are not strictly monotonic across all investability proxies, and each proxy is highly collinear with log market capitalisation ($r > 0.5$ for both coverage and institutional ownership), preventing separate identification in multivariate specifications. The paper therefore treats investability as a composite channel proxied primarily by size, with coverage and ownership providing corroborative descriptive evidence rather than independent causal leverage.

Governance variables do not emerge as a robust third pricing channel. Board independence share and CEO-chair separation are individually and jointly insignificant in the main E1 horse-race specifications. Appendix regressions show isolated significant coefficients at specific events (e.g., fresh independence at E3), but these do not form a consistent pattern across specifications or events. Director-feed coverage is rich (≈260/274 E1 firms), so the binding constraint is measurement scope rather than sample size: our data cover board *structure* (independence share, CEO-chair separation, busy-board indicators, and board size) rather than board *quality*. Governance dimensions not captured in our data—disclosure quality, shareholder engagement, or internal audit effectiveness—might predict reform CARs, and their absence limits the paper's ability to rule out governance as a channel.

The watch-list result, while economically large and directionally consistent across two events (E1: +5.2% under FF3, +3.2% under CAPM in Table 8; E4: +12.7% under FF3), rests on a treatment group of only five firms. The E4 regression coefficient (+11.1% under FF3, +8.2% under CAPM) does not achieve statistical significance at conventional levels, so the result should be treated as suggestive evidence for the stigma-relief mechanism rather than definitive identification.

Finally, post-August 2025 prices are sourced from Yahoo Finance and lack bid-ask quotes and exact traded values. Cross-sectional variables for post-E1 events are forward-filled from the most recent available snapshot, except market capitalisation and liquidity, which are recomputed from contemporaneous prices. These data constraints apply primarily to E4 and E5 and do not affect the core E1 results. All findings are specific to the SGX setting and may not generalise to markets with different institutional investor bases, regulatory structures, or levels of secondary market liquidity.

7 Conclusion

Singapore's February 2025 equity market reform deployed two policy instruments under a unified mandate: government-funded capital deployment to draw institutional money into the market, and regulatory liberalisation to lower the cost of remaining listed. Both targeted firms at the

bottom of the exchange's ecosystem—small, illiquid, thinly covered, and regulatory-burdened companies. The market priced both instruments as likely to miss those firms, and each shortfall traces to the institutional channels through which the reform was transmitted.

Capital deployment was priced as bypassing smaller firms because of intermediation. Professional fund managers operate within an institutional architecture—fiduciary obligations, AUM-based compensation, liquidity requirements, benchmark-relative evaluation—optimised for deploying capital into large, liquid securities. Directing S$5 billion through these intermediaries and expecting it to reach micro-cap equities is equivalent to routing water through pipes that narrow to a pinhole: the constraint is in the plumbing, not the pressure. The market understood this at E1, pricing the reform as a large-cap windfall. The intermediation gap is not a failure of fund managers, who responded rationally to the incentives they face, but a structural property of the transmission mechanism. When, five months later, three specific managers were named with mandates explicitly requiring significant small- and mid-cap allocation, the size gradient reversed. Intermediated capital deployment can work, but its effectiveness depends on the specificity and credibility of mandate terms that constrain the intermediary's natural portfolio tilt.

Deregulation was priced as information loss rather than burden relief. The reform's architects viewed SGX's public oversight apparatus—the query regime, the financial watch-list, the trading halt framework—as a cost imposed on listed firms. The market viewed it as information infrastructure. Trading halts and suspensions told investors which firms had serious compliance problems; routine regulatory queries, by contrast, were not priced as informative. The market's reliance on accumulated regulatory history, rather than recent scrutiny, further indicates that oversight functioned as long-term reputational capital. Dismantling these mechanisms did not resolve the underlying problems they tracked; it proposed to make those problems opaque. The market accordingly penalised firms that had attracted the most regulatory attention. The penalty, however, was not uniform. For financially distressed firms trapped in a self-reinforcing cycle of watch-list inclusion, financing impairment, and low valuation, the regulatory machinery itself was part of the problem. Removing the watch-list removed the trap, and the market priced this targeted relief positively at both announcement and implementation. Broad oversight removal was penalised when it appeared to eliminate useful monitoring; targeted watch-list abolition was rewarded where it removed an identified stigma mechanism whose costs exceeded its informational value.

The broader lesson is that well-intentioned reform fails when it ignores the structural constraints of its transmission channels. This problem is not unique to Singapore. Across developed markets, the growth of passive investment, the expansion of private equity, and the dominance of US equities in global portfolio allocation have systematically redirected capital away from smaller listed firms. Government programmes designed to counteract these trends face a common structural problem: the professional intermediaries through which public market capital must flow are themselves shaped by the institutional architecture that produced the capital deficit. This self-referential quality—where the channels through which corrective capital flows are themselves products of the structural biases being corrected—is documented in the theoretical literature on benchmarking and delegated management and diagnosed at the policy level by successive UK reviews of institutional investment. The European Union's Listing Act (European Parliament and Council of the European Union, 2024), Japan's GPIF allocation shifts, and Korea's Corporate Value-Up Programme all face

variants of the same constraint. The gap is partly a rational response to real risk differentials: small listed firms fail more frequently and present acute information asymmetries. But it operates too broadly, screening out viable small- and mid-cap firms alongside genuinely distressed ones. The policy challenge is not to override intermediary caution wholesale, but to design mandate terms—as the E3 announcement demonstrated—that narrow the gap without forcing capital into unviable positions.

The reform's final event (E5) suggests that market structure reforms which directly address the liquidity deficit—rather than channelling capital through it—can benefit illiquid firms even when the overall event favours larger ones, though the bundling of market structure enhancements with further EQDP capital and the dual listing bridge at E5 prevents clean identification of the market structure channel in isolation.

The two channels may also have interacted at E1. Because small firms disproportionately carry regulatory friction, the negative signal from proposed deregulation—that oversight of problematic firms would be weakened—may have amplified the intermediation gap by reinforcing negative sentiment toward small caps as a whole. The cross-sectional correlation between size and friction ($r \approx -0.21$) is suggestive: the deregulatory component of the bundled reform may have actively worsened the capital deployment component's size tilt, making the two failures not merely parallel but self-reinforcing. Formal testing of this interaction remains an avenue for future work.

The intermediation gap and the friction penalty are two instances of the same phenomenon: the distance between regulatory intent and market outcome, mediated by institutional structure. Both forms of mismatch can be mitigated through more precise design: specific, credible mandate terms for capital deployment, and targeted removal of identified burden mechanisms rather than wholesale dismantling of oversight. The reform's own timeline illustrates this logic. When generic intent was replaced by specific commitment at E3, the market revised. When the watch-list was abolished at E4, the positive response concentrated among previously watch-listed firms, albeit in a treatment group too small for statistical precision. The distance between good policy and good outcomes is, in the end, a design problem.

References

George A. Akerlof. The market for "lemons": Quality uncertainty and the market mechanism. *Quarterly Journal of Economics*, 84(3):488–500, 1970. 2, 6

Yakov Amihud. Illiquidity and stock returns: cross-section and time-series effects. *Journal of Financial Markets*, 5(1):31–56, 2002. 15

Sriya Anbil. Managing stigma during a financial crisis. *Journal of Financial Economics*, 130(1): 166–181, 2018. doi: 10.1016/j.jfineco.2017.05.014. 6

Olivier Armantier, Eric Ghysels, Asani Sarkar, and Jeffrey Shrader. Discount window stigma

during the 2007–2008 financial crisis. *Journal of Financial Economics*, 118(2):317–335, 2015. doi: 10.1016/j.jfineco.2015.08.006. 6

John Armour, Colin Mayer, and Andrea Polo. Regulatory sanctions and reputational damage in financial markets. Technical report, SSRN Electronic Journal, 2010. 6

John Armour, Colin Mayer, and Andrea Polo. Naming and shaming: Evidence from event studies. *The Cambridge Handbook of Compliance*, 2019. doi: 10.1017/9781108759458.050. 6

Abhijit V. Banerjee and Esther Duflo. Do firms want to borrow more? Testing credit constraints using a directed lending program. *Review of Economic Studies*, 81(2):572–607, 2014. doi: 10.1093/restud/rdt046. 4

Vanessa Behrmann, Lars Hornuf, Daniel Vrankar, and Jochen Zimmermann. The deregulation of quarterly reporting and its effects on information asymmetry and firm value. *Review of Quantitative Finance and Accounting*, 2024. doi: 10.1007/s11156-024-01338-5. 6

James A. Bennett, Richard W. Sias, and Laura T. Starks. Greener pastures and the impact of dynamic institutional preferences. *Review of Financial Studies*, 16(4):1203–1238, 2003. doi: 10.1093/rfs/hhg040. 5

Benjamin Braun and Brett Christophers. Asset manager capitalism: An introduction to its political economy and economic geography. *Environment and Planning A: Economy and Space*, 56(2): 546–557, 2024. 4

Andrea Buffa, Dimitri Vayanos, and Paul Woolley. Asset management contracts and equilibrium prices. Working paper, 2014. 5

Brian J. Bushee and Christian Leuz. Economic consequences of sec disclosure regulation: Evidence from the otc bulletin board. *Journal of Accounting and Economics*, 2004. doi: 10.1016/j.jacceco.2004.04.002. 6

Cécile Carpentier, Douglas Cumming, and Jean-Marc Suret. The value of capital market regulation: Ipos versus reverse mergers. *Journal of Empirical Legal Studies*, 2010. doi: 10.1111/j.1740-1461.2011.01247.x. 6

Hans B. Christensen, Luzi Hail, and Christian Leuz. Mandatory ifrs reporting and changes in enforcement. *Journal of Accounting and Economics*, 2013. doi: 10.1016/j.jacceco.2013.10.007. 6

Hans B. Christensen, Luzi Hail, and Christian Leuz. Capital-market effects of securities regulation: Prior conditions, implementation, and enforcement. *Review of Financial Studies*, 29(11):2885–2924, 2016. doi: 10.1093/rfs/hhw055. 6

Stijn Claessens, Daniela Klingebiel, and Sergio L. Schmukler. Stock market development and internationalization: Do economic fundamentals spur both similarly? *Journal of Empirical Finance*, 13(3):316–350, 2006. doi: 10.1016/j.jempfin.2006.03.002. 7

John C. Coffee. Law and the market: The impact of enforcement. Technical report, SSRN Electronic Journal, 2007. 6

John C. Coffee, Jr. Market failure and the economic case for a mandatory disclosure system. *Virginia Law Review*, 70(4):717–803, 1984. doi: 10.2307/1073083. 6

Maria M. Correia. Political connections and SEC enforcement. *Journal of Accounting and Economics*, 57(2–3):241–262, 2014. doi: 10.1016/j.jacceco.2014.04.004. 6

K. J. Martijn Cremers and Antti Petajisto. How active is your fund manager? A new measure that predicts performance. *Review of Financial Studies*, 22(9):3329–3365, 2009. doi: 10.1093/rfs/hhp057. 5

Stefania D'Amico and Thomas B. King. Flow and stock effects of large-scale treasury purchases: Evidence on the importance of local supply. *Journal of Financial Economics*, 108(2):425–448, 2013. doi: 10.1016/j.jfineco.2012.11.007. 4

Dhammika Dharmapala and Vikramaditya S. Khanna. The costs and benefits of mandatory securities regulation: Evidence from market reactions to the jobs act of 2012. Technical report, SSRN Electronic Journal, 2014. 6

Tatiana Didier. Information asymmetries and institutional investor mandates. Working paper, 2011. 23

Craig Doidge, G. Andrew Karolyi, and René M. Stulz. The U.S. listing gap. *Journal of Financial Economics*, 123(3):464–487, 2017. doi: 10.1016/j.jfineco.2016.12.002. 5

European Banking Authority. Report on SMEs and SME supporting factor. Technical Report EBA/OP/2016/04, European Banking Authority, 2016. URL https://www.eba.europa.eu/sites/default/documents/files/documents/10180/1359456/602d5c61-b501-4df9-8c89-71e32ab1bf84/EBA-Op-2016-04%20%20Report%20on%20SMEs%20and%20SME%20supporting%20factor.pdf. March 2016. 5

European Parliament and Council of the European Union. Regulation (EU) 2024/2809 of 23 october 2024 amending regulations (EU) 2017/1129, (EU) no 596/2014 and (EU) no 600/2014 to make public capital markets in the Union more attractive for companies and to facilitate access to capital for small and medium-sized enterprises (listing act), 2024. URL https://eur-lex.europa.eu/eli/reg/2024/2809/oj. OJ L, 2024/2809, 14 November 2024. 5, 40

Eugene F. Fama and Kenneth R. French. Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56, 1993. doi: 10.1016/0304-405X(93)90023-5. 12, 18

Eugene F. Fama and Kenneth R. French. Size, value, and momentum in international stock returns. *Journal of Financial Economics*, 2012. doi: 10.1016/j.jfineco.2012.05.011. 12

Eugene F. Fama and Kenneth R. French. A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22, 2015. doi: 10.1016/j.jfineco.2014.10.010. 12

Nuno Fernandes, Ugur Lel, and Darius P. Miller. Escape from new york: The market impact of loosening disclosure requirements. *Journal of Financial Economics*, 2008. doi: 10.1016/j.jfineco.2008.12.010. 6

Allen Ferrell. Mandatory disclosure and stock returns: Evidence from the over-the-counter market. *The Journal of Legal Studies*, 2007. doi: 10.1086/511898. 6

Joseph E. Gagnon, Matthew Raskin, Julie Remache, and Brian P. Sack. Large-scale asset purchases by the federal reserve: Did they work? *International Journal of Central Banking*, 7(1):3–43, March 2011. 4

Xiaohui Gao, Jay R. Ritter, and Zhongyan Zhu. Where have all the IPOs gone? *Journal of Financial and Quantitative Analysis*, 48(6):1663–1692, 2013. doi: 10.1017/s0022109014000015. 5

Stuart C. Gilson. Transactions costs and capital structure choice: Evidence from financially distressed firms. *The Journal of Finance*, 52(1):161–196, 1997. doi: 10.1111/j.1540-6261.1997.tb03812.x. 6

Michael Greenstone, Paul Oyer, and Annette Vissing-Jorgensen. Mandated disclosure, stock returns, and the 1964 securities acts amendments. Working paper, 2005. 6

Gustavo Grullon, Yelena Larkin, and Roni Michaely. Are US industries becoming more concentrated? *Review of Finance*, 23(4):697–743, 2019. doi: 10.1093/rof/rfz007. 5

Aurelio Gurrea-Martínez. Strengthening the international competitiveness of capital markets: Global insights and local strategies. *Singapore Management University School of Law Research Paper No. 18/2025*, 2025. doi: 10.2139/ssrn.5456074. URL https://ssrn.com/abstract=5456074. European Corporate Governance Institute – Law Working Paper No. 876/2025. 7

Zhiguo He and Wei Xiong. Delegated asset management, investment mandates, and capital immobility. *Journal of Financial Economics*, 2008. doi: 10.1016/j.jfineco.2012.08.010. 5

Paul M. Healy and Krishna G. Palepu. Information asymmetry, corporate disclosure, and the capital markets: A review of the empirical disclosure literature. *Journal of Accounting and Economics*, 31(1–3):405–440, 2001. doi: 10.1016/S0165-4101(01)00018-0. 6

Joerg-Markus Hitz and Florian Moritz. Turning back the clock on disclosure regulation? – evidence from the termination of the quarterly reporting mandate in europe. Technical report, SSRN Electronic Journal, 2019. 6

Benjamin W. Hoffman, John L. Campbell, and Jason L. Smith. Investor perceptions of government deregulation: Evidence from section 404 of the sarbanes-oxley act. *Accounting Horizons*, 2018. doi: 10.2308/horizons-17-176. 6

Peter Iliev. The effect of SOX section 404: Costs, earnings quality, and stock prices. *The Journal of Finance*, 65(3):1163–1196, 2010. doi: 10.1111/j.1540-6261.2010.01564.x. 6

Peter Iliev, Darius P. Miller, and Lukas Roth. Uninvited u.s. investors? economic consequences of involuntary cross-listings. *Journal of Accounting Research*, 2013. doi: 10.1111/1475-679x.12035. 8

Doron Israeli, Charles M. C. Lee, and Suhas A. Sridharan. Is there a dark side to exchange traded funds? An information perspective. *Review of Accounting Studies*, 22(3):1048–1083, 2017. doi: 10.1007/s11142-017-9400-8. 5

Howell E. Jackson and Mark J. Roe. Public and private enforcement of securities laws: Resource-based evidence. *Journal of Financial Economics*, 93(2):207–238, 2009. doi: 10.1016/j.jfineco.2008.08.006. 7

Tim Jenkinson and Tarun Ramadorai. Does one size fit all? the consequences of switching markets with different regulatory standards. *European Financial Management*, 2008. doi: 10.1111/j.1468-036x.2013.12021.x. 6

Kathleen M. Kahle and René M. Stulz. Is the US public corporation in trouble? *Journal of Economic Perspectives*, 31(3):67–88, 2017. doi: 10.1257/jep.31.3.67. 5

John Kay. The Kay review of UK equity markets and long-term decision making. Technical report, Department for Business, Innovation and Skills, London, 2012. URL https://www.gov.uk/government/publications/kay-review. Final Report, July 2012. 4

V van Kervel, M Larrain, and J Sabat. Scale economies in liquidity provision: Evidence from designated market makers. Technical report, SSRN Working Paper, 2025. 11

Kenneth Khoo, Sinchit Lai, and Chuyue Tian. The impact of antitrust enforcement on china's digital platforms: Evidence from samr v. alibaba. *International Review of Law and Economics*, 83:106268, 2025. 7

Arvind Krishnamurthy and Annette Vissing-Jorgensen. The effects of quantitative easing on interest rates: Channels and implications for policy. *Brookings Papers on Economic Activity*, 2011(2):215–287, 2011. doi: 10.1353/eca.2011.0019. 4

Rafael La Porta, Florencio Lopez-de Silanes, Andrei Shleifer, and Robert W. Vishny. Legal determinants of external finance. *The Journal of Finance*, 52(3):1131–1150, 1997. doi: 10.1111/j.1540-6261.1997.tb02727.x. 5

Rafael La Porta, Florencio Lopez-de Silanes, Andrei Shleifer, and Robert W. Vishny. Law and finance. *Journal of Political Economy*, 106(6):1113–1155, 1998. doi: 10.1086/250042. 5

Rafael La Porta, Florencio Lopez-de Silanes, and Andrei Shleifer. What works in securities laws? *The Journal of Finance*, 61(1):1–32, 2006. doi: 10.1111/j.1540-6261.2006.00828.x. 5

J. Lawrence. The economics of market confidence: (ac)costing securities market regulations. *Company and Securities Law Journal*, 17:171, 2000. 4

Murphy Jun Jie Lee. *The Microstructure of Trading Processes on the Singapore Exchange*. PhD thesis, Finance Discipline Group, UTS Business School, University of Technology, Sydney, 2013. 7

Christian Leuz and Robert E. Verrecchia. The economic consequences of increased disclosure. *Journal of Accounting Research*, 38(Supplement):91–124, 2000. doi: 10.2307/2672910. 6

Ernest Lim. *A case for shareholders' fiduciary duties in common law Asia.* Cambridge University Press, 2019. 7

Ernest WK Lim. Concentrated ownership, state-owned enterprises and corporate governance. *Oxford Journal of Legal Studies*, 41(3):663–691, 2021. 7

A. Craig MacKinlay. Event studies in economics and finance. *Journal of Economic Literature*, 35(1): 13–39, 1997. 7

Paul G. Mahoney. The exchange as regulator. *Virginia Law Review*, 83(7):1453–1500, 1997. doi: 10.2307/1073764. 7

Y. T. Mak and Yuan Li. Determinants of corporate ownership and board structure: Evidence from Singapore. *Journal of Corporate Finance*, 7(3):235–256, 2001. doi: 10.1016/S0929-1199(01)00021-9. 7

Justin Marion. How costly is affirmative action? Government contracting and California's Proposition 209. *Review of Economics and Statistics*, 91(3):503–522, 2009. doi: 10.1162/rest.91.3.503. 4

Elizabeth Maynes and John Rumsey. Conducting event studies with thinly traded stocks. *Journal of Banking & Finance*, 17(1):145–157, 1993. 15

Paul Myners. Institutional investment in the united kingdom: A review. Review, HM Treasury, London, March 2001. 4

Ulf Nielsson. Do less regulated markets attract lower quality firms? evidence from the london aim market. Technical report, SSRN Electronic Journal, 2013. 6

Vincent Ooi and Cheng Han Tan. Singapore company law and the economy: reciprocal influence over 50 years. *Asia Pacific Law Review*, 27(1):14–38, 2019. 7

Tim C. Opler and Sheridan Titman. Financial distress and corporate performance. *The Journal of Finance*, 49(3):1015–1040, 1994. doi: 10.1111/j.1540-6261.1994.tb00086.x. 6

Sam Peltzman. Toward a more general theory of regulation. *The Journal of Law and Economics*, 19 (2):211–240, 1976. doi: 10.1086/466865. 5

Dan W. Puchniak and Luh Luh Lan. Independent directors in singapore: Puzzling compliance requiring explanation†. *The American Journal of Comparative Law*, 2017. doi: 10.1093/ajcl/avx023. 7

Kurt Schmidheiny and Sebastian Siegloch. On event study designs and distributed-lag models: Equivalence, generalization and practical implications. Technical report, SSRN Electronic Journal, 2019. 7

G. William Schwert. Using financial data to measure effects of regulation. *The Journal of Law and Economics*, 24(1):121–158, 1981. doi: 10.1086/466977. 7

Andrei Shleifer and Daniel Wolfenzon. Investor protection and equity markets. *Journal of Financial Economics*, 66(1):3–27, 2002. doi: 10.1016/S0304-405X(02)00149-6. 5

Roger Silvers. The valuation impact of sec enforcement actions on nontarget foreign firms: The valuation impact of sec enforcement actions. *Journal of Accounting Research*, 2016. doi: 10.1111/1475-679x.12098. 6

George J. Stigler. The theory of economic regulation. *The Bell Journal of Economics and Management Science*, 2(1):3–21, 1971. doi: 10.2307/3003160. 5

Cheng Han Tan, Dan W Puchniak, and Umakanth Varottil. State-owned enterprises in singapore: Historical insights into a potential model for reform. *Colum. J. Asian L.*, 28:61, 2014. 7

Hans Tjio, Yingxin Hu, and Poh Chen Ong. *Principles and Practice of Securities Regulation in Singapore*. LexisNexis, Singapore, 3rd edition, 2017. URL https://store.lexisnexis.com/en-sg/principles-and-practice-of-securities-regulations-in-singapore-3rd-edition.html. 7

David Walker. A review of corporate governance in UK banks and other financial industry entities. Review, HM Treasury, London, November 2009. 4

Wai Yee Wan, Christopher Chen, and Say H. Goo. Public and private enforcement of corporate and securities laws: An empirical comparison of hong kong and singapore. Technical report, European Business Organization Law Review, 2017. 7

Jeffrey Wurgler. On the economic consequences of index-linked investing. In William T. Allen, Rakesh Khurana, Jay Lorsch, and Gerald Rosenfeld, editors, *Challenges to Business in the Twenty-First Century: The Way Forward*. American Academy of Arts and Sciences, 2010. doi: 10.3386/w16376. 5

P Tan Tjin Yi. The design of market admission rules: Insights from 3 asian alternative markets. Technical report, SSRN Working Paper, 2024. 7

Ivy Xiying Zhang. Economic consequences of the Sarbanes–Oxley act of 2002. *Journal of Accounting and Economics*, 44(1–2):74–115, 2007. doi: 10.1016/j.jacceco.2007.02.002. 6

A Robustness and Supplementary Results

This appendix collects identification tests, specification robustness checks, supplementary figures, and additional event-level and cross-sectional results.

A.1 Identification and Placebo Tests

A.1.1 Paired Reversal Placebo

The E1-to-E3 reversal in the size coefficient is central to the intermediation gap hypothesis. To assess whether such a reversal could arise by chance, we construct a paired-date reversal placebo. Two hundred pairs of pseudo-event dates are sampled uniformly from June 2023 to January 2025, each pair separated by approximately 105 trading days (matching the E1–E3 gap) and excluding the window around E0. For each pair, firm-level CARs are estimated at both dates using a standard $[-250, -30]$ estimation window, and the size and friction coefficients from the horse race specification in Equation (13) are computed. The reversal statistic $\Delta\hat{\beta} = \hat{\beta}_{\text{date}_2} - \hat{\beta}_{\text{date}_1}$ measures the change in each coefficient across the pair.

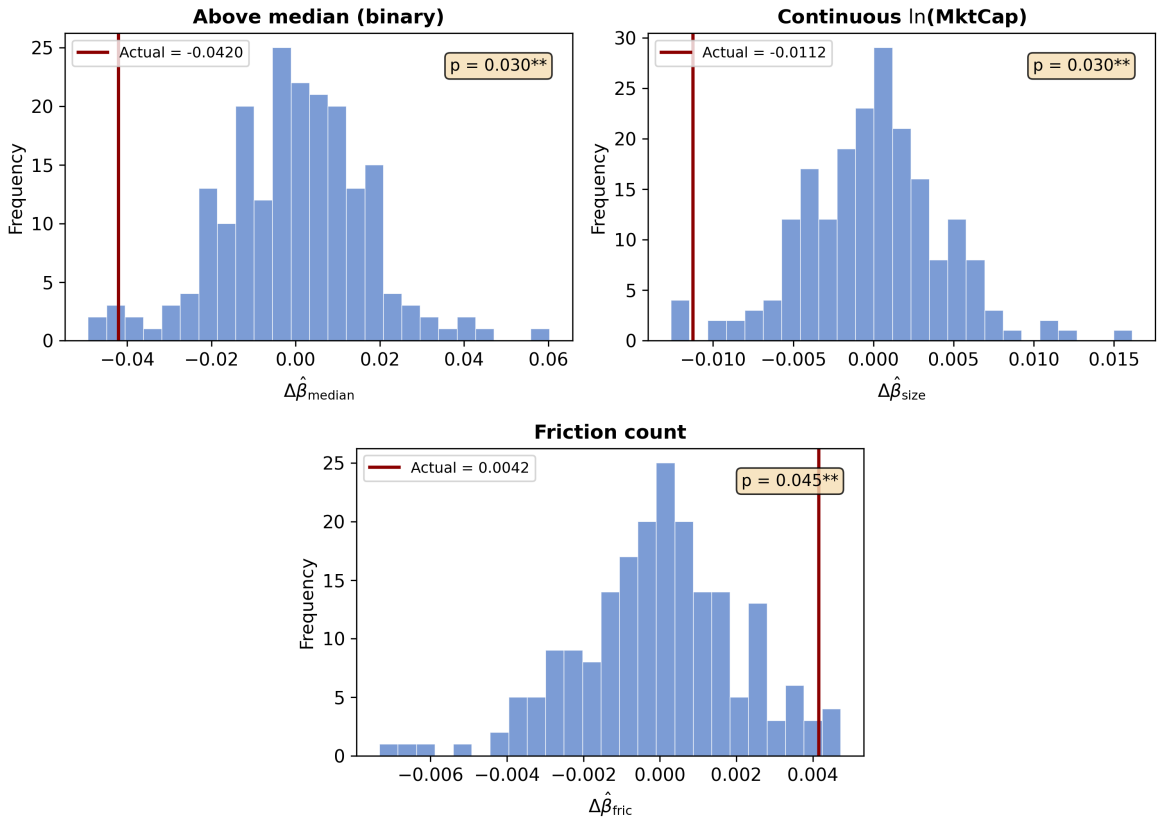
Table 16 reports the results. The binary above-median indicator produces the most extreme reversal: $\Delta\hat{\beta}_{\text{median}} = -0.042$ falls at the 2nd percentile of the placebo distribution ($p = 0.030$), with only 6 of the 200 placebo pairs generating a reversal of comparable magnitude. The continuous size measure reversal ($\Delta\hat{\beta}_{\text{size}} = -0.011$, 2nd percentile, $p = 0.030$) is likewise unusual. Both size reversals reject the null that the E1-to-E3 sign flip could arise from random cross-sectional variation. The friction coefficient reversal ($\Delta\hat{\beta}_{\text{fric}} = +0.004$, 98th percentile, $p = 0.045$) is also significant at the 5% level, confirming that the information channel exhibits an unusual cross-event pattern alongside the intermediation channel. Figure 6 plots the placebo distributions with the actual reversals marked.

Table 16: Paired Reversal Placebo (200 Pseudo-Event Pairs)

Variable	Actual $\Delta\hat{\beta}$	Placebo Mean	Placebo SD	Pctile	Perm. p
Above median	−0.0420	−0.0007	0.0169	2.0%	0.030^{**}
log(Market cap)	−0.0112	−0.0001	0.0044	2.0%	0.030^{**}
Friction count	+0.0042	−0.0000	0.0021	98.0%	0.045^{**}

Notes: 200 pseudo-event date-pairs separated by ≈105 trading days (matching the E1–E3 gap). $\Delta\hat{\beta} = \hat{\beta}_{\text{date}_2} - \hat{\beta}_{\text{date}_1}$ from the horse race specification (Equation (13)) with HC1 robust standard errors. Two-sided p: fraction of placebos with $|\Delta\hat{\beta}| \geq |\Delta\hat{\beta}_{\text{actual}}|$. Above median = 1 if $\ln(\text{MktCap})_i \geq$ sample median. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 6: Distribution of placebo coefficient reversals from 200 random date-pairs. Vertical lines mark the observed E1-to-E3 reversals.



Note: Top-left panel: distribution of $\Delta\hat{\beta}_{\text{median}}$ (binary above-median coefficient) across 200 placebo pairs; the actual reversal (−0.0420, maroon line) falls at the 2nd percentile ($p = 0.030$). Top-right panel: distribution of $\Delta\hat{\beta}_{\text{size}}$ (continuous ln(Market cap) coefficient); the actual reversal (−0.0112) falls at the 2nd percentile ($p = 0.030$). Bottom panel: distribution of $\Delta\hat{\beta}_{\text{fric}}$ (friction count); the actual reversal (+0.0042) falls at the 98th percentile ($p = 0.045$). All three reversals are statistically significant at the 5% level.

A.1.2 Cross-Market Placebo: Hong Kong

This subsection reports the full regression results underlying the cross-market placebo summarised in Figure 5. Each SGX firm is matched to its three nearest HKEX neighbours by log market capitalisation ($K = 3$), producing a matched sample of 644 Hong Kong firms. The identical size–CAR specification is then estimated on both markets at E1. Table 17 Panel A reports results for both the continuous $\ln(\text{MktCap})_i$ specification and the binary above-median indicator. Under the continuous measure, the SGX coefficient is positive and significant (+0.006, $p < 0.01$) while the HK coefficient is negative and marginally significant (−0.004, $p < 0.10$); the cross-market difference is significant ($p = 0.001$). The binary indicator produces an even sharper contrast ($p < 0.001$). Panel B reports size quartile CARs within each market: the SGX gradient increases monotonically from $Q1$ to $Q4$, while the HK sample exhibits a significantly negative gradient, with a $Q4-Q1$ spread of −2.78 percentage points ($p = 0.024$), opposite in sign to the SGX effect.

Table 17: Cross-Market Placebo: SGX vs Matched Hong Kong Sample (E1)

	SGX	HK (matched)	Difference	p
Panel A: Size regressions (E1, FF3 CAR$[-1,+3]$)				
log(Market cap)	$+0.0057^{***}$	-0.0040^{*}	$+0.0097^{***}$	0.001
	(0.0020)	(0.0021)	(0.0029)	
Above median	$+0.0248^{***}$	-0.0166^{*}	$+0.0414^{***}$	0.000
	(0.0071)	(0.0088)	(0.0113)	
N	272	644	916	
R^2	0.035	0.005		

Panel B: Size quartile CARs (%, E1)

	SGX		HK (matched)	
	Mean CAR	N	Mean CAR	N
Q1 (smallest)	-2.75^{***}	68	-0.41	161
Q2	-2.33^{***}	68	-1.96^{**}	161
Q3	-0.34	68	-2.51^{***}	161
Q4 (largest)	$+0.22$	68	-3.19^{***}	161
$Q4-Q1$	$+2.98^{***}$		-2.78^{**}	
	$[t = 2.74]$		$[t = -2.26]$	
	$(p = 0.007)$		$(p = 0.024)$	
Cross-market $Q4-Q1$ difference: $+5.75$ pp*** $(p = 0.000)$				

Note: Panel A reports OLS regressions of FF3 CARs $(-1,+3)$ on size measures. The HK sample consists of 644 HKEX firms matched to SGX firms by log market capitalisation ($K = 3$ nearest neighbours). HC1 robust standard errors in parentheses. The difference column tests H_0: $\beta_{\text{SGX}} = \beta_{\text{HK}}$ using $z = (\hat{\beta}_{\text{SGX}} - \hat{\beta}_{\text{HK}})/\sqrt{SE^2_{\text{SGX}} + SE^2_{\text{HK}}}$. Panel B reports mean CARs by size quartile within each market. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.2 Specification Robustness

A.2.1 CAR Window Robustness

Table 18 reports the E1 horse race specification across five CAR windows. The above-median size indicator is significant at all five windows ($p < 0.05$), ranging from $+0.011$ at $(-1,+1)$ to $+0.025$ at $(-1,+5)$. This uniformity, combined with the wider confidence intervals around the continuous $\ln(\text{MktCap})_i$ measure reported in Section A.2.4, supports the threshold interpretation: the size effect reflects a binary investability distinction rather than a monotonic gradient. The friction coefficient is likewise robust across windows: -0.0031 ($p < 0.05$) at the baseline $(-1,+3)$, -0.0034 ($p < 0.05$) at $(-1,+5)$, and -0.0027 ($p < 0.05$) at $(0,+2)$. The narrowest window $(-1,+1)$ produces a smaller friction coefficient (-0.0017, $p < 0.05$), consistent with friction-related information being

incorporated over multiple trading days.

Table 18: CAR Window Robustness: E1 Horse Race

	$(-1,+1)$	$(-1,+3)$	$(-1,+5)$	$(0,+2)$	$(0,+5)$
Above median	0.0112**	0.0175***	0.0250***	0.0112**	0.0214**
	(0.0052)	(0.0067)	(0.0089)	(0.0051)	(0.0083)
Friction count	-0.0017**	-0.0031**	-0.0034**	-0.0027**	-0.0024*
	(0.0008)	(0.0012)	(0.0016)	(0.0011)	(0.0015)
Watchlist	0.0282	0.0512	0.1111**	0.0610*	0.0914**
	(0.0192)	(0.0327)	(0.0485)	(0.0325)	(0.0444)
Illiquidity	0.0023	-0.0018	-0.0076	-0.0088	-0.0088
	(0.0038)	(0.0098)	(0.0112)	(0.0100)	(0.0118)
Catalist	-0.0182*	-0.0098	-0.0063	-0.0028	-0.0012
	(0.0100)	(0.0126)	(0.0141)	(0.0094)	(0.0124)
N	267	272	274	269	274
R^2	0.099	0.099	0.101	0.113	0.076

Notes: OLS with HC1 robust standard errors in parentheses. The dependent variable is the firm-level cumulative abnormal return over the indicated trading-day window around E1, estimated using a Fama–French three-factor model with Asia Pacific ex Japan factors. Above median = 1 if $\ln(\text{MktCap})_i \geq$ sample median. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.2.2 Factor Model Robustness

Table 19 reports the E1 horse race specification estimated separately under CAPM, FF3, and FF5. The above-median coefficient is significant under all three models: +0.020 ($p < 0.01$) under CAPM, +0.018 ($p < 0.01$) under FF3, and +0.020 ($p < 0.05$) under FF5. The friction coefficient is stable and negative: -0.0036 ($p < 0.01$) under CAPM, -0.0031 ($p < 0.05$) under FF3, and -0.0029 ($p < 0.05$) under FF5. CAPM friction coefficients are approximately 16% larger than their FF3 counterparts, consistent with the Asia Pacific ex Japan SMB factor absorbing part of the friction effect; this reflects expected channel overlap between size and friction, not a specification weakness. The watch-list coefficient is significant under CAPM (+0.030, $p < 0.05$) but attenuated under FF3 (+0.051, insignificant) and FF5 (-0.012, insignificant), again reflecting SMB absorption. Overall, the core size and friction results are robust to factor model choice.

Table 19: Factor Model Robustness: E1 Horse Race

	CAPM	FF3	FF5
Above median	0.0204***	0.0175***	0.0198**
	(0.0064)	(0.0067)	(0.0086)
Friction count	-0.0036***	-0.0031**	-0.0029**
	(0.0011)	(0.0012)	(0.0013)
Watchlist	0.0297**	0.0512	-0.0122
	(0.0137)	(0.0327)	(0.0318)
Illiquidity	-0.0015	-0.0018	0.0281***
	(0.0030)	(0.0098)	(0.0059)
Catalist	-0.0124	-0.0098	-0.0081
	(0.0115)	(0.0126)	(0.0162)
N	272	272	234
R^2	0.143	0.099	0.116

Notes: OLS with HC1 robust standard errors in parentheses. The dependent variable is the firm-level cumulative abnormal return over the $(-1, +3)$ trading-day window around E1. CAPM: market model with SGX equal-weighted index. FF3: Fama–French three-factor model (Asia Pacific ex Japan, primary specification). FF5: Fama–French five-factor model (Asia Pacific ex Japan). The CAPM friction coefficient is ≈16% larger than FF3 because the APXJ SMB factor absorbs part of the friction–size overlap. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.2.3 Size Effect Across Alternative Measures

Panel A of Table 20 reports univariate size regressions by event: the E1 coefficient is +0.0054 ($p < 0.01$) and the E3 coefficient is -0.0059 ($p < 0.05$), a sign flip consistent with Bayesian learning as the market updated once mandate details were disclosed. Panel B confirms that the reversal is robust across four alternative size measures (log market cap, log assets, above-median indicator, and the composite z-score), all of which exhibit the E1-positive/E3-negative pattern.

Table 20: Size Effect Across Alternative Measures

	E1	E2	E3
Panel A: Univariate			
log_mktcap	0.0054***	-0.0012	-0.0059**
	(0.0018)	(0.0018)	(0.0026)
N	272	273	269
R^2	0.038	0.002	0.024

Panel B: Alternative measures (E1 vs. E3 sign flip)

	E1 β	E1 p	E3 β	E3 p	Sign Flip
log_mktcap	+0.00540	(0.003)	−0.00585	(0.024)	Yes
log_assets	+0.00309	(0.074)	−0.00542	(0.036)	Yes
above_median	+0.02492	(0.000)	−0.01751	(0.074)	Yes
top_decile					
z_size	−0.00980	(0.018)	+0.01523	(0.013)	Yes

Notes: Panel A: OLS with HC1 robust standard errors in parentheses. The dependent variable is the firm-level cumulative abnormal return over the $(-1, +3)$ trading-day window around the event date, estimated using a Fama–French three-factor model with Asia Pacific ex Japan factors. Panel B: Univariate regressions per measure. z_size sign reversed (higher = smaller). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.2.4 Continuous Size Measure

Table 21 reports the horse race specification using the continuous $\ln(\text{MktCap})_i$ measure in place of the binary above-median indicator. The size coefficient enters positively in all specifications but with wider confidence intervals than the binary measure: +0.0032 ($p < 0.10$) in the joint specification and +0.0035 ($p < 0.05$) in the governance-augmented specification. The wider standard errors reflect the concavity of the size–CAR relationship documented in Figure 3: the gradient is steep below the median and flat above it, favouring a threshold specification over a log-linear one. All other coefficients are virtually unchanged between the continuous and binary specifications.

Table 21: Horse Race with Continuous Size Measure (E1)

	(1)	(2)	(3)	(4)	(5)
log(Market cap)	0.0054***		0.0032*	0.0036	0.0035**
	(0.0018)		(0.0017)	(0.0023)	(0.0017)
Friction count		-0.0034***	-0.0032***	-0.0032**	-0.0031**
		(0.0013)	(0.0012)	(0.0015)	(0.0013)
Watchlist			0.0505	0.0560	0.0498
			(0.0327)	(0.0390)	(0.0330)
Illiquidity (z-Amihud)			-0.0017	-0.0014	-0.0010
			(0.0095)	(0.0107)	(0.0095)
Catalist			-0.0114	-0.0018	-0.0074
			(0.0124)	(0.0179)	(0.0127)
Independence share					-0.0184
					(0.0249)
No CEO duality					0.0091
					(0.0101)
Sector FE	No	No	No	Yes	No
N	272	274	271	243	260
R^2	0.038	0.049	0.087	0.195	0.083

Notes: OLS with HC1 robust standard errors in parentheses. The dependent variable is the firm-level cumulative abnormal return over the $(-1, +3)$ trading-day window around E1 (February 21, 2025), estimated using a Fama–French three-factor model with Asia Pacific ex Japan factors. Same specification as Table 13 but using continuous $\ln(\text{MktCap})_i$ instead of the binary above-median indicator. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.2.5 Catalist Board-Tier Effect

The Catalist board-tier coefficient reported in preliminary analyses (-3.0 percentage points, $p < 0.001$) was inflated by a sample composition artefact. The initial sample included non-equity stock codes (warrants, structured products, daily leverage certificates), all classified as Mainboard, whose positive mean CARs at E1 inflated the Mainboard reference group and exaggerated the apparent Catalist underperformance. Table 22 reports the equity-only results. Under CAPM, the bare Catalist coefficient is -2.82 percentage points ($p < 0.01$, Panel A), but the FF3 model, which includes the SMB factor, attenuates it to -2.44 percentage points ($p < 0.05$). Adding the above-median size indicator further reduces the coefficient to -1.23 percentage points (insignificant). Catalist firms are substantially smaller than Mainboard firms on average, confirming that the board-tier variable proxies for size rather than capturing a distinct regulatory channel.

Table 22: Catalist Board-Tier Effect

Panel A: Catalist coefficient by model and controls (E1)

	FF3		Market Model	
	Bare	+Size	Bare	+Size
Catalist	-0.0244**	-0.0123	-0.0282***	-0.0137
	(0.0121)	(0.0132)	(0.0106)	(0.0117)
Above median		0.0214***		0.0257***
		(0.0071)		(0.0068)
N	274	274	274	274
R^2	0.025	0.057	0.037	0.087

Panel B: Board means by event (FF3 CAR$(-1,+3)$ %)

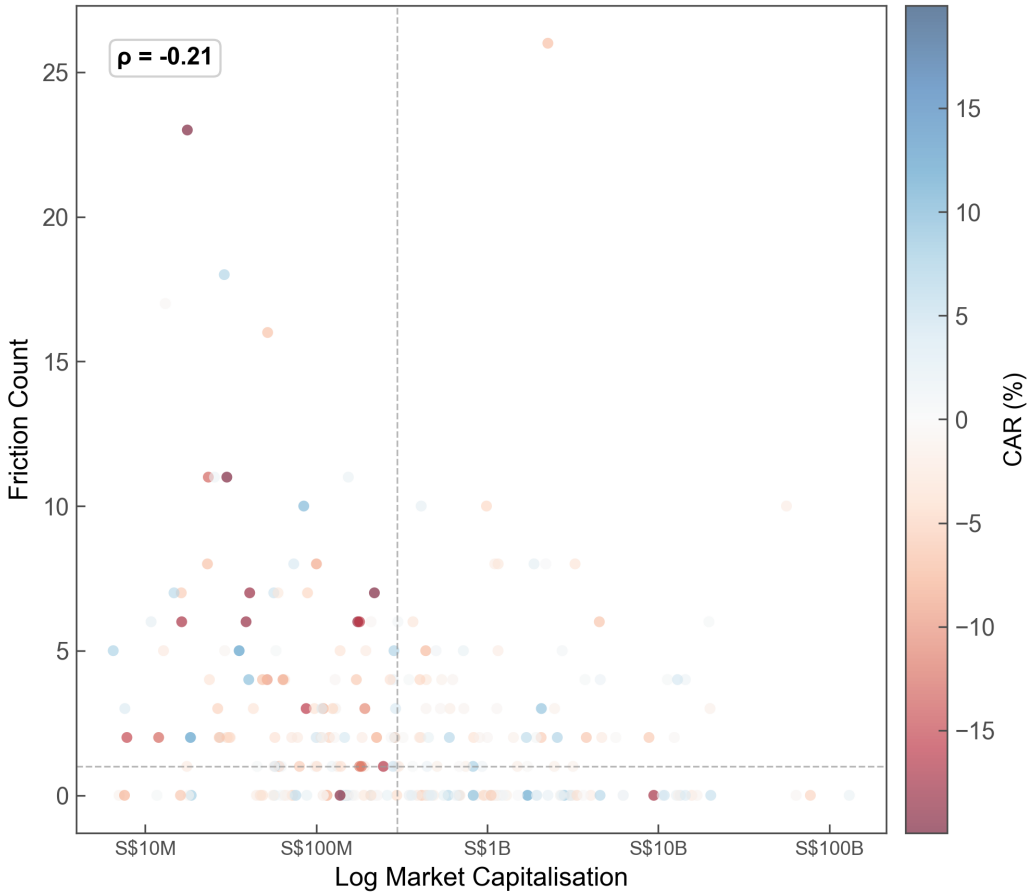
	E1	E2	E3
Mainboard	-0.95	-0.23	+2.65
Catalist	-3.39	+0.55	+5.61
Diff (Cat−MB)	-2.44*	+0.77	+2.96*

Notes: Panel A: OLS with HC1 robust standard errors in parentheses. The dependent variable is the firm-level cumulative abnormal return over the $(-1,+3)$ trading-day window around E1. FF3: Fama–French three-factor model (Asia Pacific ex Japan). Market Model: CAPM with SGX equal-weighted index. The FF3 size factor (SMB) absorbs most of the board effect. Panel B: Mean CARs (%) by board tier (equity-only sample). Diff tested via Welch t-test. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.2.6 Size–Friction Correlation

Figure 7 plots log market capitalisation against the regulatory friction count. The moderate negative correlation ($\rho = -0.21$) confirms that larger firms tend to accumulate fewer friction events, consistent with the expected overlap between the intermediation and information channels. Crucially, the correlation is far from collinear, preserving the independent explanatory power of both variables in the horse-race regressions of Table 13.

Figure 7: Scatter plot of log market capitalisation against regulatory friction count. The negative correlation (Pearson $\rho = -0.21$) confirms that size and friction are empirically related but not collinear.

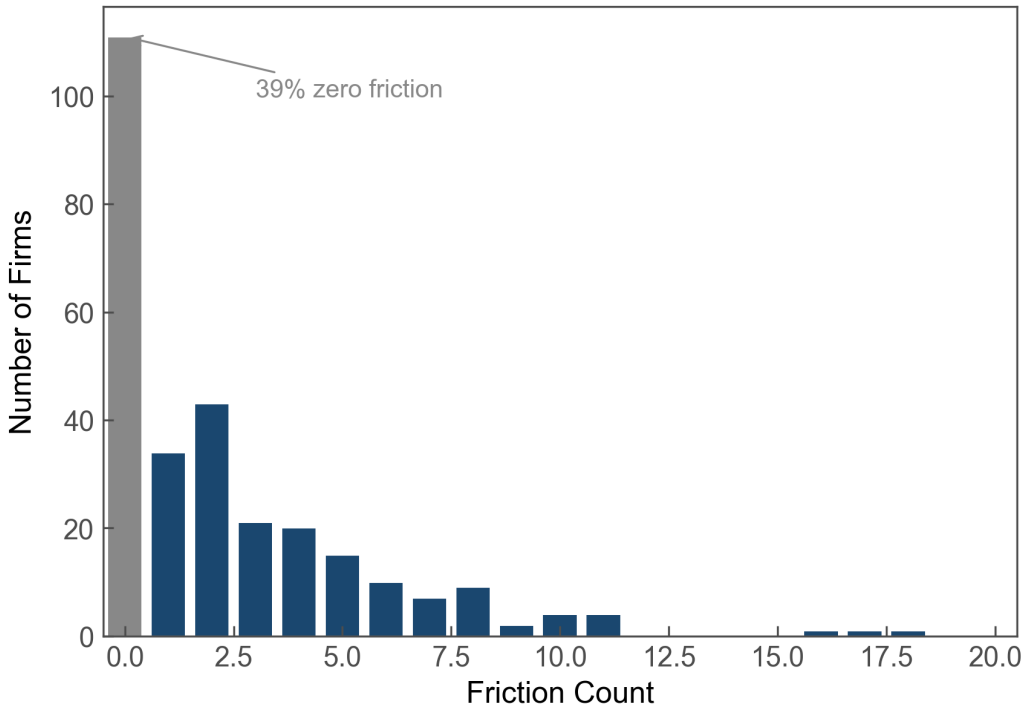


Note: Each point represents a firm at E1. Colour indicates FF3 CAR$(-1, +3)$, winsorised at the 1st and 99th percentiles. Dashed lines mark sample medians. ρ denotes the Pearson correlation between log market capitalisation and total friction count ($\rho = -0.21$), confirming that the two channels are not collinear.

A.3 Supplementary Figures

A.3.1 Friction Distribution

Figure 8: Distribution of regulatory friction counts across the SGX equity panel. The majority of firms have zero or one friction event; the right tail is driven by repeat watch-list and trading-halt episodes.



Note: Distribution of total friction event count across the 335 equity-only sample firms. Friction events include SGX queries, trading halts, suspensions, and watch-list inclusions over August 2022 to August 2025.

Figure 8 shows the heavily right-skewed friction distribution underlying the estimates in Table 8, confirming that the linear specification is driven primarily by the contrast between zero-friction and moderate-friction firms.

A.3.2 Market Index Comparison

Figure 9: Equal-weighted SGX equity index versus the Straits Times Index (STI) around the E1 announcement. The divergence reflects the large-cap tilt of the STI.

Note: Cumulative daily equal-weighted return across all SGX-listed equities with valid price observations. Vertical dashed lines mark event dates.

Figure 9 illustrates why benchmark choice matters: the STI's capitalisation weighting masks the negative average response visible in the equal-weighted index, consistent with the size effect documented in Section 5.

A.4 Additional Event and Cross-Sectional Results

A.4.1 E0: Review Group Formation

The Review Group formation announcement on August 2, 2024 (E0) produced a strongly negative mean CAR of -2.15% ($p < 0.001$, $N = 260$), with only 30.8% of firms posting positive returns. Cross-sectional regressions yield significant size and friction coefficients, consistent with the patterns documented for E1. However, the event window coincides with the late-July/early-August 2024 JPY carry trade unwind, which generated large negative returns across Asian equity markets. Augmenting the market model with a JPY factor (FF4-JPY) causes the REIT effect and several other cross-sectional patterns to disappear, indicating that the E0 results are confounded by the carry trade unwinding. We therefore treat E0 as uninformative for hypothesis testing and rely on E1 as the primary announcement event.

A.4.2 E2 and E3 Detailed Results

The E2 consultation event (May 15, 2025) produced a null mean CAR of −0.11% ($p = 0.721$, $N = 275$). No cross-sectional variable achieves significance at conventional levels, and the horse race R^2 is 0.8%. This null result is consistent with the consultation having been fully anticipated following E1's announcement that detailed proposals would follow "by mid-2025."

E3 (July 21, 2025) produced a uniformly positive mean CAR of +3.12% ($p < 0.001$, $N = 270$, 70.0% positive). Unlike E1, the continuous size measure reverses sign (−0.006, $p < 0.05$), while friction count is economically zero ($p > 0.55$), and the horse race R^2 is 4.3%. The capital deployment was a rising tide that lifted all equities, but with smaller firms benefiting disproportionately. The reversal of the size gradient at E3, combined with its positive sign at E1, constitutes the key evidence for the intermediation gap hypothesis: once specific mandate terms requiring small- and mid-cap allocation accompanied the capital deployment, the market reversed its size-based differentiation.

A.4.3 E5: Final Report

The final report event (E5, November 19, 2025) produced a positive mean CAR of +0.62% ($t = 2.76$, $p = 0.006$, $N = 258$), with 60.5% of firms posting positive returns. In the cross-section (Table 15), the above-median size indicator is positive and significant (+0.015, $p < 0.01$), and illiquidity is positively signed (+0.007, $p < 0.05$)—the opposite direction from E1 (see Section 5.5 for the tercile-stratified analysis). Friction count and watch-list status are insignificant.

A.4.4 Governance Variables

Table 23 reports univariate governance regressions across E1, E2, E3, and E4. Board independence share, non-busy board membership, and turnover rate are insignificant at all four events. CEO-chair separation is insignificant at E1, E2, and E4, but marginally significant at E3 (−0.022, $p < 0.10$). Fresh independence—defined as whether the board appointed at least one new independent director in the preceding fiscal year—is strongly positive at E3 (+0.092, $p < 0.01$) and at E4 (+0.019, $p < 0.05$), suggesting that firms with recently refreshed boards benefited disproportionately from the capital deployment and deregulation implementation.

Table 23: Governance Variables Across Events

	E1	E2	E3	E4
Independence share	0.0038	-0.0240	-0.0113	-0.0131
	(0.0260)	(0.0243)	(0.0375)	(0.0374)
No CEO duality	0.0108	0.0087	-0.0222*	0.0015
	(0.0104)	(0.0092)	(0.0126)	(0.0096)
Non-busy board	-0.0264	-0.0016	-0.0469	-0.0194
	(0.0255)	(0.0252)	(0.0409)	(0.0307)
Fresh independence	-0.0131	0.0019	0.0919***	0.0192**
	(0.0101)	(0.0125)	(0.0113)	(0.0081)
Turnover rate	-0.0046	-0.0020	-0.0019	-0.0002
	(0.0029)	(0.0026)	(0.0053)	(0.0030)

Notes: Univariate OLS with HC1 robust standard errors in parentheses. The dependent variable is the firm-level cumulative abnormal return over the $(-1, +3)$ trading-day window, estimated using a Fama–French three-factor model with Asia Pacific ex Japan factors. Governance variables from director data. E4 governance merged from E1 cross-section. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.4.5 Broker/Institutional Ownership

Singapore's nominee and custody structure means that substantial shareholdings are held through intermediary accounts. We decompose the top-20 shareholder register into four categories: insider (directors and substantial shareholders), nominee (bank nominee accounts), broker (retail securities brokers), and institutional (fund managers and sovereign wealth funds). While nominee accounts hold shares on behalf of any investor type, broker accounts predominantly serve retail investors who maintain securities accounts with their brokerage. We use *Broker %* (the share of top-20 holdings registered through retail securities brokers) as a proxy for retail-dominated ownership and, inversely, for the absence of institutional infrastructure that facilitates price discovery around information events.

Table 24 Panel A reports stepwise OLS regressions of E1 abnormal returns on broker ownership. Univariately, a one percentage point increase in broker ownership is associated with a -0.0007 lower CAR ($p < 0.01$; column 1). Adding size controls attenuates the coefficient to -0.0006 ($p < 0.05$; column 2), and further adding friction controls yields -0.0005 ($p < 0.05$; column 3). In the most saturated specification with financial controls, illiquidity, and Catalist (column 5), the coefficient remains at -0.0005 ($p < 0.10$), suggesting a genuine intermediation channel operating independently of firm size and market frictions.

Panel B examines whether post-reform ownership changes align with the policy's intended direction. If the programme channelled capital toward small and mid-cap firms, nominee hold-

ings (the primary custodial vehicle for institutional investment) should increase disproportionately for smaller firms. The observed pattern runs contrary to this prediction. Controlling for FY2024 nominee levels, larger firms exhibit higher FY2025 nominee ownership (coefficient on log(MktCap) = 1.25, $p < 0.05$; column 6), while Catalist-listed firms show a 3.5 percentage point decline in broker holdings relative to Mainboard peers ($p < 0.05$; column 7). These ownership patterns are consistent with the announcement-return evidence suggesting that intermediation favoured larger, more liquid firms, though they do not isolate the causal effect of the EQDP on ownership structure.

The broker ownership subsample covers 205 of 274 E1 firms (75%), with excluded firms being marginally smaller ($p = 0.082$) and overwhelmingly Catalist-listed (97% vs 81%, $p < 0.001$). CARs and friction counts do not differ significantly between subsamples. The ownership rebalancing sample covers 172 firms with top-20 shareholder data in both FY2024 and FY2025; results should be interpreted with these coverage caveats in mind.

Table 24: Broker Ownership and E1 Abnormal Returns

	(1)	(2)	(3)	(4)	(5)
Panel A: Broker ownership and E1 abnormal returns					
Broker %	-0.0007^{***}	-0.0006^{**}	-0.0005^{**}	-0.0006^{**}	-0.0005^{*}
	(0.0002)	(0.0003)	(0.0002)	(0.0002)	(0.0003)
Log(Mktcap)		0.0036	0.0023	0.0027	0.0011
		(0.0023)	(0.0022)	(0.0022)	(0.0026)
Friction count			-0.0033^{*}	-0.0036^{**}	-0.0037^{**}
			(0.0018)	(0.0017)	(0.0018)
Watchlist				0.0699^{*}	0.0628
				(0.0385)	(0.0419)
Illiquidity					0.0027
					(0.0095)
Catalist					−0.0057
					(0.0139)
Financial controls	No	No	No	No	Yes
N	205	204	204	204	192
R^2	0.046	0.059	0.094	0.115	0.147

Panel A (cont'd): Broker ownership quartile sorts (%)

	E1: February 2025		E3: July 2025	
	Mean	Median	Mean	Median
Q1 (lowest)	−0.36	−0.12	$+1.97^{**}$	+0.76
Q2	−0.45	−0.54	$+3.16^{***}$	+1.91
Q3	-1.75^{*}	−1.52	$+4.98^{***}$	+3.24
Q4 (highest)	-2.37^{***}	−1.94	$+3.68^{***}$	+2.85
$Q4 - Q1$	-2.00^{*}		+1.70	
N	205		201	

Panel B: Post-E3 ownership rebalancing

	(6)	(7)	(8)
Nominee % (FY2024)	0.8369^{***}		
	(0.0537)		
Log(Mktcap)	1.2515^{**}	−0.2459	
	(0.5484)	(0.2659)	
Catalist	0.2180	-3.5228^{**}	
	(1.6345)	(1.5889)	
Medium			−1.0171
		62	(1.6969)
Large			−2.8088
			(2.0851)